Daily Drilling Report

Geothermal Resource Group, Inc.

Well ID: Forge 78B-32 — **Well Name: 78B-32**

Field: UTAHFORGE — Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 2 — **Report For 28-Jun-21**

Operator:	University of Utah	**Rig:**	Frontier Rig 16	**Spud Date:**	28-Jun-21	**Daily Cost / Mud ($):**			---
Measured Depth (ft):	421	**Last Casing:**	16.000 at 416	**Wellbore:**	Original Wellbore	**AFE No.**		**AFE ($)**	**Actual ($)**
Vertical Depth (ft):		**Next Casing:**	11.750 at 3,300	**RKB Elevation (ft):**	30.40	---		---	---
Proposed TD (ft):	9500	**Last BOP Test:**	28-Jun-21	**Job Reference RKB (ft):**		---		---	---
Hole Made (ft) / Hrs:	293 / 4.0	**Next BOP Test:**	19-Jul-21	**Working Interest:**		**Totals:**		---	---
Average ROP (ft/hr):	73.25					**Well Cost ($):**			---

Drilling Days (act./plan):	2/21	**Flat Days (act./plan):**	0/9	**Total Days (act./plan):**	2/30	**Days On Location:**	2

Pers/Hrs:	**Operator:**	3 / 36	**Contractor:**	14 / 168	**Service:**	4 / 48	**Other:**	3 / 36	**Total:** 24 / 288

Safety Summary: No incidents or events reported. Conducted BOP Test, Crown Check, Safety Meeting.

Current Operations: Testing out choke manifold at report time.

Planned Operations:
Test casing along with BOPE.
Pick up 8” drill collars.
Trip in the hole with tri-cone bit.
Drill out shoe track and 10 ft of new formation to perform LOT
Pull out of the hole and lay down tricone bit
Make up 14 3/4” BHA with PDC and continue drilling 14 3/4" section.

Toolpusher: Steve Caldwell , Justin Bristol

Wellsite Supervisors: Virgil Welch , Brian Gresham — **Tel No.:** 7132807438

Operations Summary

From	To	Elapsed	End MD(ft)	Code	Operations Description	Non-Prod
0:00	1:00	1.00	128	3-34-3	Make up 22" surface hole assembly.	
1:00	1:30	0.50	128	03-051	Filled pipe and hole.	
1:30	5:00	3.50	421	03-021	Drilled 22" hole from 128' to 421'.	
5:00	5:30	0.50	421	03-051	Circulated well bore clean.	
5:30	6:00	0.50	421	3-27	Wiped hole f/ 421' t/ 100' with no issues. No fill on bottom.	
6:00	6:30	0.50	421	3-6-2	Pull out of the hole f/ 421'	
6:30	7:00	0.50	421	3-34-4	Lay down NMDC and bit	
7:00	8:00	1.00	421	4-56	Rig up casing crew.	
8:00	10:00	2.00	421	4-12-1	Rigged up and Ran 9 joints of 16" k-55 64# BTC casing (382.89') of total casing. Landed out with 33' landing joint. Shoe set at 416' KB. Float collar at 372' KB.	
10:00	11:00	1.00	421	4-56	Rig down casing crew	
11:00	12:30	1.50	421	4-12-1	Land casing with well head 3' below ground level.	
12:30	13:00	0.50	421	5-56	RIH with stab-in and stab into float collar at 378	
13:00	14:30	1.50	421	5-56	Rig up cement crew	
14:30	16:00	1.50	421	5-99	Pressure test to 1500 psi, Pump 20 bbl fresh water, pump 50 bbls of sepiolite, pump 5 bbl of fresh water, pump 13 bbl od sodium silkicate, pump 5 bbls fresh water, Mix and pump 146 bbl of 13.88 RC Theremelite w/ 2% CC, displace w/ 6 bbls fresh water, check floats CIP @ 16:00	
16:00	16:30	0.50	421	5-56	Rig down cement crew	
16:30	17:00	0.50	421	5-6-2	POH with stab-in	
17:00	0:00	7.00	421	1-14	Installed BOP as followed. 16” x 20” 2M DSA, 20” 2M spacer spool, 20” 2M x 20” 3M DSA, 20” 3M mud cross, 20” 3M X 20” 2M DSA, 20” 2M double gates, 20” 2M annular, 20” 2M flow spool.	

Management Summary

Drilled 22" hole from 128' to 421'. Circulated hole clean. Wiped hole from 421’ to 100’.
Pulled out laying down tools.
Rigged up and ran 9 joints of 65 ppf, K-55 16” Buttress casing setting shoe at 416’ KB.
Ran in the hole with stab-in tool.
Rigged up Resource cementing and cemented casing in place CIP @ 16:00.
Rigged down Resource cementing.
Pulled out of the hole with stab-in tool.
Installed 20” BOPE. Nippled up BOPE.

Comments

Good cement to surface.

Daily Drilling Report

Geothermal Resource Group, Inc.

Well ID: Forge 78B-32 — **Well Name: 78B-32**

Field: UTAHFORGE — Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 2 — **Report For 28-Jun-21**

Casing/Tubular Information

Type	Size (ins)	Top MD (ft)	Top TVD (ft)	Bottom MD (ft)	Bottom TVD (ft)	Hole Section	OH Diam. (ins)	Nom. Wgt. (lbs)	Nominal Grade	LOT (lbs/gal)
FULL	16.000	0		416		SURF	22.000	65	K-55	

Mud Information

Dens.	Vis	PV	YP	Filt.	Cake	pH/ES	% Solids	% Oil	% Water	% Sand	% LGS	Cl	Ca	CaCl	Gels 10s	Gels 10m	Gels 30m	Temp In	Temp Out	Mud Loss
28-Jun-21 05:15 at Depth 421 ft Mud Pits																				
8.90	51	19	22	12.2	1	9	11	0	89	2.5		800	120	0	9	16	25	85	85	10.0

Bit/BHA/Workstring Information

No	Run	Make	Model	Diam	Depth In	This Run Dist	This Run Hrs	R.O.P. Avg	R.O.P. Max	WOB	RPM	Torque	Mud Wt	Flow	Pump Press	J. Vel	P. Drp	HHP	JIF
1	1	Smith	XR-C	22.000	128	293	3.5	83.7	250.0	15	90	7	9	800	576	142	161	75	524

Jets: 28 28 28 **Out:** 421 **Grade: Cutter:** 1 / 1 **Dull:** NO / NO **Wear:** A **Brgs:** 0 **Gge:** 0 **Pull:** TD

BHA - No. 1 - BIT, BS, XO, MWD, XO, MONEL, FLOAT, OTHER, XO, HWDP = 314.30

Drilling Parameters

Depth (ft) From	Depth (ft) To	ROP (ft/hr) Avg	ROP (ft/hr) Max	WOB (lbs) Avg	WOB (lbs) Max	RPM Avg	RPM Max	Torque (ft lbs) Avg	Torque (ft lbs) Max	Flow (gals/min) Avg	Flow (gals/min) Max	Pressure (psi)
128	421	80.0	120.0	15	30	75	80	3	8	650	800	1,100

Annular Velocity: Drill Collars: 37.9 **Drill Pipe:** 34.7

Miscellaneous Drilling Parameters

Hook Loads (lbs): Off Bottom Rotate: 85 **Pick Up:** 100 **Slack Off:** 78 **Drag Avg/Max:** 3 / 5

Hours on BHA: Since Inspection: 3.5 **Total:** **Jars:**

Survey Information

Survey Type	Meas. Depth	Inc.	Azimuth	TVD	Closure	Vertical Section	Coordinates N-S	Coordinates E-W	D.L.S.
MWD	241.0	0.69	358.3	241.0	1.5	1.5	N 1.5	W 0.0	0.286
MWD	337.0	1.04	359.5	337.0	2.9	2.9	N 2.9	W 0.1	0.365

Rig Information

Equipment Problems: None

Location Condition: Good.

Transport:

Solids Control Information

Screen Sizes:	Top	Middle 1	Middle 2	Bottom
Shaker No 1:	60	60	60	60
Shaker No 2:	60	60	60	60
Shaker No 3:	60	60	60	60

Equipment Usage (Hrs):

Desander: 4 **Desilter:** 4 **Degasser:** 0

Centrifuge 1: 4 (Solids Removal)

Bulk Inventory

Item Type	Units	Beginning	Used	Received	On Hand	Net
Diesel (OBM)		7,637	940		6,697	

Drill Pipe Inventory

DP Size	Joints	Weight	Grade	Thread	DP Size	Joints	Weight	Grade	Thread
5"	114	19.5	S-135	4.5IF					

Safety Information

Meetings/Drills	Time	Description
Safety	60	Running casing, cementing casing, Nipple up, Pinch points

First Aid Treatments: 0 **Medical Treatments:** 0 **Lost Time Incidents:** 0 **Days Since LTI:**

☑ **BOP Test** ☑ **Crownamatic Check**

Daily Drilling Report

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 2	**Report For 28-Jun-21**

Weather Information

Sky Condition:	Clear	**Visibility:**	10
Air Temperature:	66 degF	**Bar. Pressure:**	29.96
Wind Speed/Dir:	8 / NNE	**Wind Gusts:**	2

Daily Drilling Report

Geothermal Resource Group, Inc.

Well ID: Forge 78B-32 | **Well Name: 78B-32**

Field: UTAHFORGE | Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 4 | **Report For 30-Jun-21**

Operator:	University of Utah	**Rig:**	Frontier Rig 16	**Spud Date:**	28-Jun-21	**Daily Cost / Mud ($):**	---
Measured Depth (ft):	2699	**Last Casing:**	16.000 at 416	**Wellbore:**	Original Wellbore	**AFE No.** / **AFE ($)** / **Actual ($)**	
Vertical Depth (ft):	2699	**Next Casing:**	11.750 at 3,300	**RKB Elevation (ft):**	30.40	--- / --- / ---	
Proposed TD (ft):	9500	**Last BOP Test:**	29-Jun-21	**Job Reference RKB (ft):**		--- / --- / ---	
Hole Made (ft) / Hrs:	2,266 / 15.0	**Next BOP Test:**	19-Jul-21	**Working Interest:**		**Totals:** --- / ---	
Average ROP (ft/hr):	151.07			**LOT (lbs/gal):**	0.58	**Well Cost ($):**	---
Drilling Days (act./plan):	4/21	**Flat Days (act./plan):**	0/9	**Total Days (act./plan):**	4/30	**Days On Location:**	4

Pers/Hrs:	**Operator:**	3 / 36	**Contractor:**	14 / 168	**Service:**	4 / 48	**Other:**	3 / 36	**Total:** 24 / 288

Safety Summary: No incidents or events reported. Conducted Safety Meeting.

Current Operations: Pulled out of the hole from 951' to Surface.
Changed bit. Tipped in the hole to 2,699'.
Drilled 14 3/4" hole section from 2,699' to 2,818' (report time)

Planned Operations: Drill 14 3/4" section to casing point.

Toolpusher: Steve Caldwell , Justin Bristol

Wellsite Supervisors: Virgil Welch , Brian Gresham | **Tel No.:** 7132807438

Operations Summary

From	To	Elapsed	End MD(ft)	Code	Operations Description	Non-Prod
0:00	3:30	3.50	433	3-34-3	Picked up 14 3/4" RSS BHA to 433'.	
3:30	4:15	0.75	665	3-2-3	Drilled 14 3/4" hole from 433' to 665' with full returns.	
4:15	4:30	0.25	665	03-051	Circulated hole clean to rack back HWDP and place 3 stands of 8" drill collars into BHA.	
4:30	5:30	1.00	665	3-34-3	Pulled out of the hole from 665' to 389'. Racked back 3 stands of HWDP. Installed 3 stands of 8" drill collars into the BHA. Tripped in the hole to 665'.	
5:30	6:00	0.50	858	3-2-3	Drilled 14 3/4" hole from 665' to 858 with full returns.	
6:00	10:00	4.00	2,077	3-2-3	Drilled 14-3/4" hole from 858' to 2,077' with full returns.	
10:00	10:30	0.50	2,077	SERV	Serviced rig / torqued saver sub	
10:30	11:00	0.50	2,232	3-2-3	Drilled 14-3/4" hole from 2,077' to 2,232' with full returns.	
11:00	11:30	0.50	2,232	REPR	Rig blacked out/ cleaned #2 generator radiator due to overheating.	X
11:30	12:30	1.00	2,359	3-2-3	Drilled 14-3/4" hole from 2,232' to 2,359' with full returns.	
12:30	13:30	1.00	2,359	REPR	Rig Repair cleaned #1 generator radiator due to overheating.	X
13:30	14:00	0.50	2,399	3-2-3	Drilled 14-3/4" hole from 2,359' to 2,399' with full returns.	
14:00	15:30	1.50	2,399	REPR	Rig Repairs cleaned #3 generator radiator due to overheating.	X
15:30	21:15	5.75	2,399	3-2-3	Drilled 14-3/4" hole from 2,399' to 2,699' with full returns.	
21:15	22:00	0.75	2,399	3-5-1	Circulate hole clean. Pump 15 bbl slug.	
22:00	0:00	2.00	2,399	3-6-4	Pull out of the hole from 2,699' to 951'. Tight hole encountered from 2,351' to 1,799' 50-100k over string weight. Extended trip time due to tight connections and using rig tongs. Removed rotating rubber at 850'.	

Management Summary

Picked up 14 3/4" RSS BHA to 433'.
Drilled 14 3/4" hole from 433' to 665' with full returns.
Circulated hole clean.
Pulled out of the hole from 665' to 389'.
Racked back 3 stands of HWDP.
Installed 3 stands of 8" drill collars into the BHA.
Tripped in the hole to 665'.
Drilled 14 3/4" hole from 665' to 2,077' with full returns. Serviced rig.
Drilled 14-3/4" hole from 2,077' to 2,232' with full returns. Circulated while cleaning radiator on gen #2 due to overheating.
Drilled 14-3/4" hole from 2,232' to 2,359' with full returns. Circulated while cleaning radiator on gen #1 due to overheating.
Drilled 14-3/4" hole from 2,359' to 2,399' with full returns. Circulated while cleaning radiator on gen #3 due to overheating.
Drilled 14-3/4" hole from 2,399' to 2,699' with full returns. Circulated hole clean. Tripped out of the hole to 951'.

Comments

BOPE test results emailed DWR (Jim Goddard).

Casing/Tubular Information

Type	Size (ins)	Top MD (ft)	Top TVD (ft)	Bottom MD (ft)	Bottom TVD (ft)	Hole Section	OH Diam. (ins)	Nom. Wgt. (lbs)	Nominal Grade	LOT (lbs/gal)
FULL	16.000	0		416		SURF	22.000	65	K-55	0.58

Daily Drilling Report

Geothermal Resource Group, Inc.

Well ID: Forge 78B-32 | **Well Name: 78B-32**

Field: UTAHFORGE | Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 4 | **Report For 30-Jun-21**

Mud Information

Dens.	Vis	PV	YP	Filt.	Cake	pH/ES	% Solids	Oil	Water	Sand	LGS	Cl	Ca	CaCl	Gels 10s	Gels 10m	Gels 30m	Temp In	Temp Out	Mud Loss
30-Jun-21 21:00 at Depth 2,697 ft Mud Pits, Type: Low Solids Non-Dispersed																				
9.10	42	14	13	9.4	1	10	9	0	91	12.5	4.2	800	120		3	7	11	110	118	

Mud Consumables

Item Description	Qty.	Cost	Item Description	Qty.	Cost
Barite - 100#SK	20	---	Desco CF - 25#SK	1	---
DMA - 50#SK	5	---	Drispac LV - 50#SK	5	---
Engineering - EACH	1	---	FlowZan - 25#SK	10	---
Gel - 100#SK	70	---	Micro C - 50#SK	50	---
pallets - OTHER	13	---	perdiem - OTHER	1	---
Trucking - EACH	1	---			

Bit/BHA/Workstring Information

No	Run	Make	Model	Diam	Depth In	This Run Dist	This Run Hrs	R.O.P. Avg	R.O.P. Max	WOB	RPM	Torque	Mud Wt	Flow	Pump Press	J. Vel	P. Drp	HHP	JIF
2	1	BAKER	GT-C1	14.750	360	166	3	55.3	1,	40	80	15	9	950	3050	409	1361	754	1829
Jets: 18 18 18					**Out:** 433	**Grade:**	**Cutter:**	1 / 1		**Dull:** NO / NO			**Wear:** A	**Brgs:** 0	**Gge:**			**Pull:** BHA	
3	1	NOV	TKC66	14.750	433	2,266	15	151.1					9	950	3050	281	645	357	1259
Jets: 15 15 15 15 16 16					**Out:** 2699	**Grade:**	**Cutter:**	3 / 1		**Dull:** LT / CT			**Wear:** N	**Brgs:** X	**Gge:**			**Pull:** PR	

BHA - No. 3 - BIT, OTHER, STAB, MWD, 4 OTHER, MMTR, RR, FLOAT, 2 OTHER, DC, XO, HWDP = 956.61

Drilling Parameters

Depth (ft) From	Depth (ft) To	ROP (ft/hr) Avg	ROP (ft/hr) Max	WOB (lbs) Avg	WOB (lbs) Max	RPM Avg	RPM Max	Torque (ft lbs) Avg	Torque (ft lbs) Max	Flow (gals/min) Avg	Flow (gals/min) Max	Pressure (psi)
433	2,699	174.3	1,300.0	40	65	70	120	13	22	950	1,050	3,150

Annular Velocity: **Drill Collars:** 197.7 **Drill Pipe:** 128.2

Miscellaneous Drilling Parameters

Hook Loads (lbs): **Off Bottom Rotate:** 125 **Pick Up:** 150 **Slack Off:** 100 **Drag Avg/Max:** 10 / 15

Slow Circulation Data:

Pump 1:	35 spm	95 psi	45 spm	210 psi	60 spm	375 psi
Pump 2:	35 spm	95 psi	45 spm	210 psi	60 spm	375 psi
Pump 3:	35 spm	95 psi	45 spm	210 psi	60 spm	375 psi

Hours on BHA: **Since Inspection:** 9 **Total:** 9 **Jars:** 0

Hours on Casing/Liner: **Rotating:** 9 / 0 **Tripping:** 3 / 0 ☐ **Wear Bushing Installed**

Survey Information

Survey Type	Meas. Depth	Inc.	Azimuth	TVD	Closure	Vertical Section	Coordinates N-S	Coordinates E-W	D.L.S.
MWD	545.0	0.17	93.9	545.0	4.8	4.8	N 4.8	E 0.2	0.513
MWD	731.0	0.07	271.8	731.0	4.8	4.8	N 4.8	E 0.4	0.129
MWD	822.0	0.12	37.4	822.0	4.8	4.8	N 4.8	E 0.4	0.187
MWD	914.0	0.15	342.6	914.0	5.0	5.0	N 5.0	E 0.4	0.138
MWD	1,005.0	0.47	187.3	1,005.0	4.8	4.8	N 4.8	E 0.3	0.670
MWD	1,101.0	0.19	350.9	1,101.0	4.5	4.5	N 4.5	E 0.3	0.682

Daily Drilling Report

Geothermal Resource Group, Inc.

Well ID: Forge 78B-32 | **Well Name: 78B-32**

Field: UTAHFORGE | Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 4 | **Report For 30-Jun-21**

Mud Log Information

Depth (ft) From	Depth (ft) To	TVD (ft) From	TVD (ft) To	Gas (Units) Avg	Gas (Units) Max	at Depth	Gas Connect.	Gas Trip	Drilling Exp.	Pore Press	Mud Dens.	Shale Dens.	ROP Shale	ROP Sand
0	430	0	430	0	0						9.10			
Formation Name:														
Lithology:	100% Alluvium													
430	2,430	430	2,430	0	0									
Formation Name:														
Lithology:	100% Alluvium													
2,440	2,630	2,430	2,440	0	0									
Formation Name:	60% Clay, 40% Alluvium													
Lithology:	90% Clay 10% Alluvium													
2,630	2,650	2,440	2,570	0	0									
Formation Name:	90% Clay 10% Alluvium													
Lithology:	100% Alluvium													
2,650	2,670	2,570	2,580	0	0									
Formation Name:	70% Clay, 30% Rhyolite													
Lithology:	10-50% Rhyolite 50-90% Alluvium													
2,670	2,700	2,580	2,640	0	0									
Formation Name:	90$ Rhyolite, 10% Clay													
Lithology:	100% Alluvium													

Rig Information

Equipment Problems: Generators overheating. cleaned radiators on all 3 gens.

Location Condition: Good.

Transport:

Solids Control Information

Screen Sizes:	Top	Middle 1	Middle 2	Bottom
Shaker No 1:	60	60	60	60
Shaker No 2:	60	60	60	60
Shaker No 3:	60	60	60	60

Bulk Inventory

Item Type	Units	Beginning	Used	Received	On Hand	Net
Rig Fuel		6,220	1,624	3,500	8,096	1,876

Drill Pipe Inventory

DP Size	Joints	Weight	Grade	Thread	DP Size	Joints	Weight	Grade	Thread
5"	114	19.5	S-135	4.5IF					

Safety Information

Meetings/Drills	Time	Description
Safety	60	Two 30 min Pre tour safety meetings with both crews. Trip hazards, pinch points, Working at heights.

First Aid Treatments: 0 | **Medical Treatments:** 0 | **Lost Time Incidents:** 0 | **Days Since LTI:**

☐ **BOP Test** ☐ **Crownamatic Check**

Weather Information

Sky Condition:	Partly Cloudy	**Visibility:**	10
Air Temperature:	75 degF	**Bar. Pressure:**	29.94
Wind Speed/Dir:	10 / W	**Wind Gusts:**	2

Daily Drilling Report

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 5 **Report For 01-Jul-21**

Drill Pipe Inventory

DP Size	Joints	Weight	Grade	Thread	DP Size	Joints	Weight	Grade	Thread
5"	114	19.5	S-135	4.5IF					

Safety Information

Meetings/Drills	Time	Description
Safety	60	Two 30 min Pre tour safety meetings with both crews. Handling BHA, Tripping pipe, Rigging up and running casing.

First Aid Treatments:	0	**Medical Treatments:**	0	**Lost Time Incidents:**	0	**Days Since LTI:**	

- [ ] **BOP Test**
- [x] **Crownamatic Check**

Weather Information

Sky Condition:	Clear	**Visibility:**	10
Air Temperature:	70 degF	**Bar. Pressure:**	29.83
Wind Speed/Dir:	6 / NNW	**Wind Gusts:**	1

Daily Drilling Report

Geothermal Resource Group, Inc.

Well ID: Forge 78B-32 | **Well Name: 78B-32**

Field: UTAHFORGE | Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 19 | **Report For 15-Jul-21**

Operator:	University of Utah	**Rig:**	Frontier Rig 16	**Spud Date:**	28-Jun-21	**Daily Cost / Mud ($):**			---
Measured Depth (ft):	7613	**Last Casing:**	11.750 at 2,990	**Wellbore:**	Original Wellbore	**AFE No.**		**AFE ($)**	**Actual ($)**
Vertical Depth (ft):	7611	**Next Casing:**	7.000 at 8,500	**RKB Elevation (ft):**	30.40	---		---	---
Proposed TD (ft):	9500	**Last BOP Test:**	03-Jul-21	**Job Reference RKB (ft):**		---		---	---
Hole Made (ft) / Hrs:	0 / 9.5	**Next BOP Test:**	24-Jul-21	**Working Interest:**		**Totals:**		---	---
Average ROP (ft/hr):	0.0			**LOT (lbs/gal):**	11.70	**Well Cost ($):**			---
Drilling Days (act./plan):	19/21	**Flat Days (act./plan):**	0/9	**Total Days (act./plan):**	19/30	**Days On Location:**			19

Pers/Hrs:	**Operator:**	3 / 36	**Contractor:**	14 / 168	**Service:**	6 / 72	**Other:**	3 / 36	**Total:**	26 / 312

Safety Summary: No incidents or events reported. Conducted Crown Check, Safety Meeting.

Current Operations:
Tripped in the hole from 2,613' to 7,072'.
Filled pipe and broke circulation every 1,000'.
Took weight at 7,072'. Work string free with 90k in over-pull. Ream from 7,072' to 7,231'.

Planned Operations:
Ream from 7,231' to 7,613' as needed.
Drilled 10-5/8" hole to 8,500'.

Toolpusher: Steve Caldwell , Justin Bristol

Wellsite Supervisors: Virgil Welch , Brian Gresham | **Tel No.:** 7132807438

Operations Summary

From	To	Elapsed	End MD(ft)	Code	Operations Description	Non-Prod
0:00	8:00	8.00	7,613	REPR	Waited on hydraulic hose to arrive. Pumping down the string to cool tools every hour at 300 gpm.	X
8:00	13:00	5.00	7,613	3-6-4	POH from 6,559' to 629'	
13:00	16:30	3.50	7,613	3-6-2	Racked back HWDP along with 8' drill collars. Laid out motor, bit, stabs. Motor took a high amount of torque to drain 45 kft/lbs. Bit had 3-plugged jets with pieces of rubber from the stator bit was also DBR'ed. Motor Stabilizer was 0.5" under-gauged Stabilizer above the motor was 0.75" under-gauged. Top Stabilizer was 1.25" under-gauged.	
16:30	22:30	6.00	7,613	3-34-3	Picked up new (Bico) motor 1.5 deg, 0.160 rev/gal. Picked up new 10-5/8" NOV bit. Picked up 1- Re-Run Roller Reamers 10-3/8" stab placed at 39.53'. Picked up new 10-1/2" spiral stab placed at 100.82'. Placed cubic sensors at 50.96', 132.86'. Made up and scribed tools. Tripped in the hole with BHA to 1,000'.	
22:30	23:30	1.00	7,613	3-6-3	Tripped in the hole from 1,000' to 2,613'.	
23:30	0:00	0.50	7,613	SERV	Serviced rig. Adjusted service loop.	

Management Summary

Waited on hydraulic hose to arrive.
Installed new top-drive service link hydraulic hose.
Tripped out of the hole from 6,559' to BHA.
Racked back HWDP along with 8' drill collars.
Laid out motor, bit, stabs.
Made up 10-5/8" directional assembly.
Tripped in the hole from 1,000' to 2,613'.
Serviced rig. Adjusted service loop.

Comments

Motor took a high amount of torque to drain 45 kft/lbs. Bit had 3-plugged jets with pieces of rubber from the stator bit was also DBR'ed.
Motor Stabilizer was 0.5" under-gauged
Stabilizer above the motor was 0.75" under-gauged.
Top Stabilizer was 1.25" under-gauged.

Total hours of NPT on report 8.0 HRS
Total hours of NPT on Frontier to date 56 HRS.

Daily Drilling Report **Well ID: Forge 78B-32** Field: UTAHFORGE	**Geothermal Resource Group, Inc.** **Well Name: 78B-32** Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 28 **Report For 24-Jul-21**

Operations Summary

From	To	Elapsed	End MD(ft)	Code	Operations Description	Non-Prod
16:15	17:00	0.75	8,545	4-12-1	Run 7", 41#, T-110SS, Tenaris casing from 6,614' to 7,070'. Note: Floats failed at 6,795' while running casing gained 81k in hook load. All joints were torqued to manufactures optimum recommendation of 17,070 FT/LBS. Ran all 15-joints without filling.	
17:00	17:30	0.50	8,545	4-99	Performed continuity test on fiber optics at 7,070'. Test indicated one of the five fiber-optic had failed at 4,000' from surface. Decision was made to run in one joint to 7,106' and circulate bottoms up.	
17:30	19:00	1.50	8,545	3-5-1	Circulated bottoms up at 7,106'. Max surface flow temperature was 99 degrees Fahrenheit. Static temperature reading at 6,966' with fiber-optic cable was 359 degrees Fahrenheit.	
19:00	19:45	0.75	8,545	4-12-1	Run 7", 41#, T-110SS, Tenaris casing from 7,070' to 7,812'. Note: All joints were torqued to manufactures optimum recommendation of 17,070 FT/LBS. Ran all 15-joints without filling.	
19:45	20:00	0.25	8,545	4-99	Performed continuity test on fiber optics at 7,812'. Test indicated all five fiber-optic cables failed from 4,500' to 7,812'. Static temperature was 240 degrees Fahrenheit at 4,500'.	
20:00	21:15	1.25	8,545	4-12-1	Ran 7", 41#, T-110SS, Tenaris casing from 7,812' to set depth of 8,508'. Shoe set at 8,508' KB. Float collar at 8,414' KB. Note: All joints were torqued to manufactures optimum recommendation of 17,070 FT/LBS. Ran all 15-joints without filling. Thread locked last 3 joints in the hole below landing joint.	
21:15	21:30	0.25	8,545	4-99	Performed continuity test on fiber optics at 8,508'. Test indicated all five fiber-optic cables failed from 3,933' to 8508'. Static temperature was 208 degrees Fahrenheit at 3,933'.	
21:30	22:30	1.00	8,545	4-99	Rigged down CRT tool. Rigged up cement head. Broke circulation.	
22:30	0:00	1.50	8,545	3-5-1	Circulated at 16 bbl's per min to cool well bore. Mixed and pumped 4ppb Micro C while circulating.Cleaned and cleared floor while circulating.	

Management Summary

Ran 7", 41#, T-110SS, Tenaris casing from 2,380' to 3,722'.
Installed new spool of fiber-optic cable on unit. Ran fiber-optic cable to rig floor.
Spliced on new fiber-optic cable at 3,722'.
Ran 7", 41#, T-110SS, Tenaris casing from 3,711' to set depth of 8,508'.
Rigged down CRT tool.
Rigged up cement head.
Broke circulation.
Circulated at 16 bbl's per min to cool well bore.
Mixed and pumped 4ppb Micro C while circulating.
Cleaned and cleared floor while circulating.

Comments

Splice on new fiber-optic cable at 3,722'.
Performed continuity test on fiber optics every 15 joints while running casing.
Floats failed at 6,795' while running empty casing.
Performed continuity test on fiber optics at 7,070'.
Test indicated one of the five fiber-optic had failed at 4,000' from surface.
Performed continuity test on fiber optics at 7,812'.
Test indicated all five fiber-optic cables failed from 4,500' to 7,812'.
Static temperature was 240 degrees Fahrenheit at 4,500'.
Performed continuity test on fiber optics at 8,508'.
Test indicated all five fiber-optic cables failed from 3,933' to 8508'.
Static temperature was 208 degrees Fahrenheit at 3,933'.

Government of **Western Australia**
Department of **Mines and Petroleum**

RECORD 2014/12

GSWA HARVEY 1 WELL COMPLETION AND PRELIMINARY INTERPRETATION REPORT, SOUTHERN PERTH BASIN

by
AS Millar and J Reeve

Geological Survey of Western Australia

ROYALTIES FOR REGIONS
EXPLORATION INCENTIVE SCHEME

Government of **Western Australia**
Department of **Mines and Petroleum**

Record 2014/12

GSWA HARVEY 1 WELL COMPLETION AND PRELIMINARY INTERPRETATION REPORT, SOUTHERN PERTH BASIN

compiled by
AS Millar and J Reeve*

* Aztech Well Construction Pty Ltd, Colin Street, West Perth WA 6005

Perth 2014

Geological Survey of Western Australia

Contents

Figures

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Appendices*

* Can be found as separate attachments on the accompanying USB.

GSWA Harvey 1 well completion and preliminary interpretation report, southern Perth Basin

by

AS Millar and J Reeve*

Abstract

GSWA Harvey 1 is a stratigraphic well drilled in early 2012 within the southern Perth Basin by the Geological Survey of Western Australia (GSWA). The well is located on a structural feature known as the Harvey Ridge, a poorly defined northwesterly–southeasterly trending basement high observed in seismic data and associated with a high gravity anomaly. The well was completed as part of the South West Hub Carbon Capture and Storage Project, a project jointly funded by the State and Commonwealth Governments, and industry. State funding was from the Royalties for Regions Exploration Incentive Scheme (EIS).

Based upon earlier assessments the area was interpreted to contain a significant thickness of a saline aquifer within the lower Lesueur Sandstone (Wonnerup Member), which could act as a reservoir for storing carbon dioxide. Regionally it is overlain by a potential baffle, the upper Lesueur Sandstone (Yalgorup Member, previously known as the Myalup Member). In turn, the upper Lesueur Sandstone is overlain by the Eneabba Formation, which may also include potential baffle and seal units. Importantly, the area does not contain the Yarragadee Formation, a significant fresh groundwater aquifer in the Perth Basin which might otherwise be at risk of contamination.

Results from the L198 Lower Lesueur Seismic Survey (2011), together with existing seismic data and limited nearby drilling data, were used to plan the location and design of the well to acquire the pre-competitive data needed to fill some of the identified knowledge gaps.

Harvey 1 was spudded on 7 February 2012 and the total depth of 2945 m measured depth below rotary table (MDRT) was reached on 8 March 2012. Following wireline logging, Harvey 1 was plugged with four cement plugs and the rig was released on 26 March 2012. Broadly the predicted stratigraphy was intersected; however, depths of formation boundaries were shallower than predicted by about 120–145 m.

A full mud logging program was completed with cuttings collected at 15 m intervals from 60 to 840 m, and 5 m intervals from 840 to 2945 m. A coring program designed to obtain representative rock samples for analysis was completed and recovered six 100 mm (4") diameter cores over four intervals for a total of 217 m of recovered core. Cored intervals were lithologically logged in detail and 90 representative horizontal and vertical plugs were obtained for analysis.

Standard porosity and permeability measurements were completed on core plugs, and plug trims were used for mercury injection capillary pressure (MICP) testing, X-ray diffraction (XRD) and petrological examination. Following longitudinal cutting, core was run through the GSWA HyLogger to obtain high-resolution photographs and spectral data for mineralogical studies. To assist with sediment provenance studies, a set of unwashed samples was collected for U–Pb detrital zircon geochronology.

Both logging while drilling and end of well wireline logging were used in the well formation evaluation. A single logging suite comprising five runs was completed for the 216 mm (8 ½") hole section at the completion of drilling.

Harvey 1 encountered the expected stratigraphic sequence for the Harvey Ridge area, although each formation was intersected at a shallower depth than predicted. Initial reservoir analysis has indicated that the Wonnerup Member may be suitable for the storage of CO_2.

KEYWORDS: Jurassic, Triassic, carbon capture and storage, coring, core analysis, geochronology, organic petrology, paleosols, palynology, petrology, reservoir data, spectral analysis, well logging

* Aztech Well Construction Pty Ltd, Colin Street, West Perth WA 6005

Introduction

GSWA Harvey 1 (hereafter referred to as Harvey 1) was drilled as part of the South West Hub Carbon Capture and Storage (CCS) Project, Australia's first CCS Flagship project. The South West Hub CCS Project is a government and industry partnership to develop an economically and environmentally sustainable means of reducing carbon dioxide (CO_2) emissions in the southwest region of Western Australia.

The key stakeholders of the project are as follows:

- Commonwealth Government through the Department of Industry
- WA State Government through the Department of Mines and Petroleum (DMP)
- industry through a partnership with Griffin Energy, Verve Energy, BHP Billiton Worsley Alumina, Premier Coal, Perdaman Chemicals and Fertilisers, and Alcoa Australia
- the local community of southwestern Western Australia.

Additional information on Australian CSS projects may be obtained from the Geoscience Australia (GA) website <www.ga.gov.au/scientific-topics/energy/ghg> or, for Western Australian projects, from the DMP website <www.dmp.wa.gov.au/ccs>.

The geological structural unit known as the Harvey Ridge was identified in regional studies as having potential for CO_2 storage in saline aquifers (Varma et al., 2007, 2014). The area was interpreted to contain a significant thickness of a saline aquifer within the lower Lesueur Sandstone which could act as a reservoir; it is overlain by the upper Lesueur Sandstone which is a potential baffle. The latter is overlain by the Eneabba Formation, which may also include potential baffle and seal units. Importantly, the area does not contain the Yarragadee Formation, a significant fresh groundwater aquifer in the Perth Basin which might otherwise be at risk of contamination.

Early regional studies (Causebrook et al., 2006, Varma et al., 2007, 2014) highlighted the following gaps in available data:

- There was a lack of conclusive evidence of a regional seal formation.
- There was generally poor seismic and well coverage in the area.
- There was a paucity of modern petrophysical data to determine the reservoir characteristics of the Lesueur Sandstone.

In March 2011, the Geological Survey of Western Australia (GSWA), a division of DMP, in conjunction with GA, completed a 106 km 2D seismic survey along shire roads (Fig.1). Data from the survey is available from the GA website <www.ga.gov.au/products/servlet/controller?event=GEOCAT_DETAILS&catno=74810>.

The results from the 2011 seismic survey and existing seismic data were used to plan a stratigraphic well, Harvey 1, that would provide pre-competitive data. A seismic interpretation of the Harvey area is covered by Zhan (2014). The well was designed to assist in filling some of the identified knowledge gaps, and specifically it would:

- confirm the presence of the predicted stratigraphy
- test the presence of a lower 'shale unit' in the Eneabba Formation
- enable evaluation of the sealing capacity of the shale unit, if present
- provide fresh core samples of both the Eneabba Formation and Lesueur Sandstone for testing seal capacity, reservoir properties and injectivity capacity
- provide the opportunity to acquire a comprehensive suite of modern well evaluation logs
- enable calibration of the 2011 seismic data and integration into the 3D model for the area
- assist in the planning and development of future bore holes and seismic programs for further evaluation of the area.

Harvey 1 is located on private land (Lot 1326) off Riverdale Road, Cookernup (Fig.1). It is about 80 m north of shot point 2450 on the L198 Lower Lesueur Seismic Survey line 11GA_LL2 (Fig. 2).

The well was not expected to encounter hydrocarbons, but its planned depth (3000 m) and drilling program determined that a drilling rig with full provisions for hydrocarbon exploration drilling contingencies was required. Following a public tender, MB Century was awarded the contract to drill the well using Rig 7. Well design and drilling management was completed by Aztech Well Construction Pty Ltd in conjunction with GSWA.

Harvey 1 was spudded on 7 February 2012, reached a total depth of 2945 m on 9 March, and was plugged and abandoned on 23 March 2012 (Fig. 3). All depths related to the drilling are from the rotary table (MDRT) unless noted.

Research

The main objectives of the well were to acquire data and material to further assess the technical feasibility of storing, and permanently containing up to 6.5 Mt per annum over 40 years of CO_2 in a condition referred to as a 'super critical' phase. GSWA coordinated the collection of material and data from the well and its distribution to institutions and organizations awarded contracts to examine and report on the knowledge acquired from respective projects. These ongoing studies are being coordinated by DMP and Australian National Low Emissions Coal Research and Development (ANLEC

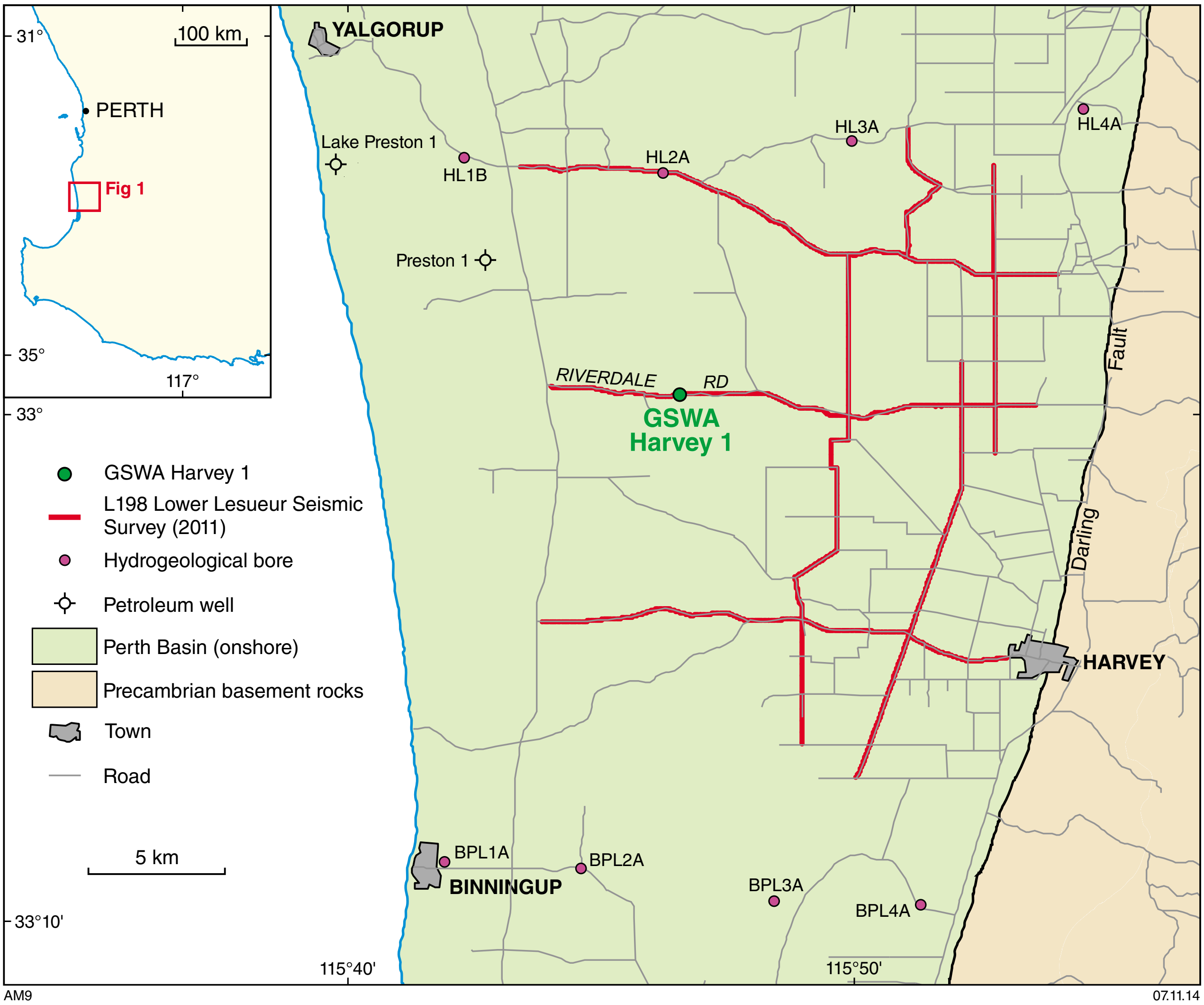

Figure 1. Location of GSWA Harvey 1, L198 Lower Lesueur Seismic Survey (2011) lines, existing petroleum wells, and hydrogeological bores

R&D), and carried out by scientists and engineers from CSIRO, Curtin University and The University of Western Australia. Projects were commissioned to evaluate the data under the following research topics:

1. Facies-based rock properties distribution
2. Geochemical characterization of gases, fluids and rocks
3. Fault seal first-order analysis
4. Advanced geophysical data analysis
5. Stratigraphic forward modelling
6. Static and dynamic reservoir modelling and simulations

Final reports on these projects are being published separately by ANLEC R&D <www.anlecrd.com.au/publications> and are also available via DMP's online database: Western Australian Petroleum and Geothermal Information Management System (WAPIMS). Results are not included within this report with the exception of preliminary core descriptions, basic core and log analysis. Additional research projects using data from Harvey 1 are ongoing and these will be made available via WAPIMS as they are completed.

Follow-up work planned for the South West Hub CCS Project includes a 3D seismic survey (2014) and several additional wells (2014–15). The objective is to advance the storage assessment from a 'pre-competitive' phase to a point where commercial considerations and decisions are made possible, which would enable the project to become a viable business opportunity for industry.

Regional geology

Harvey 1 was drilled within the southern Perth Basin on the structural feature known as the Harvey Ridge, a poorly defined, northwesterly–southeasterly trending basement high observed in seismic data and associated with a high gravity anomaly (Iasky, 1993, Crostella and Backhouse,

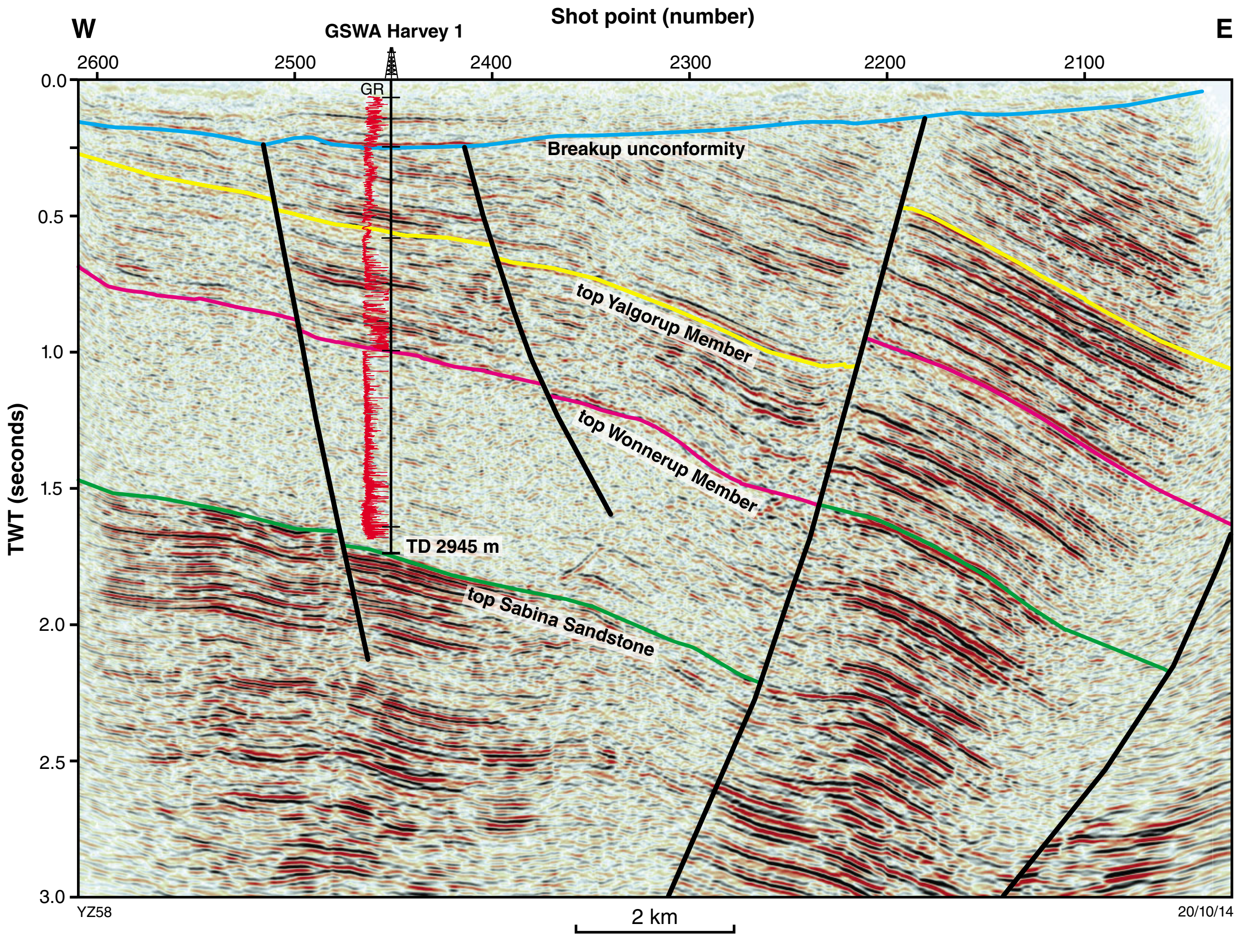

Figure 2. **Interpreted seismic section (11GA_LL2) from the L198 Lower Lesueur Seismic Survey (2011) showing the position of Harvey 1, and the stratigraphic units it intersected, including the Wonnerup and Yalgorup Members of the Lesueur Sandstone. The seismic line is located along Riverdale Road (Fig. 1). (Zhan, 2014)**

Figure 3. **MB Century Rig 7 on-site at Harvey 1 (photo courtesy of Mark Mitchell)**

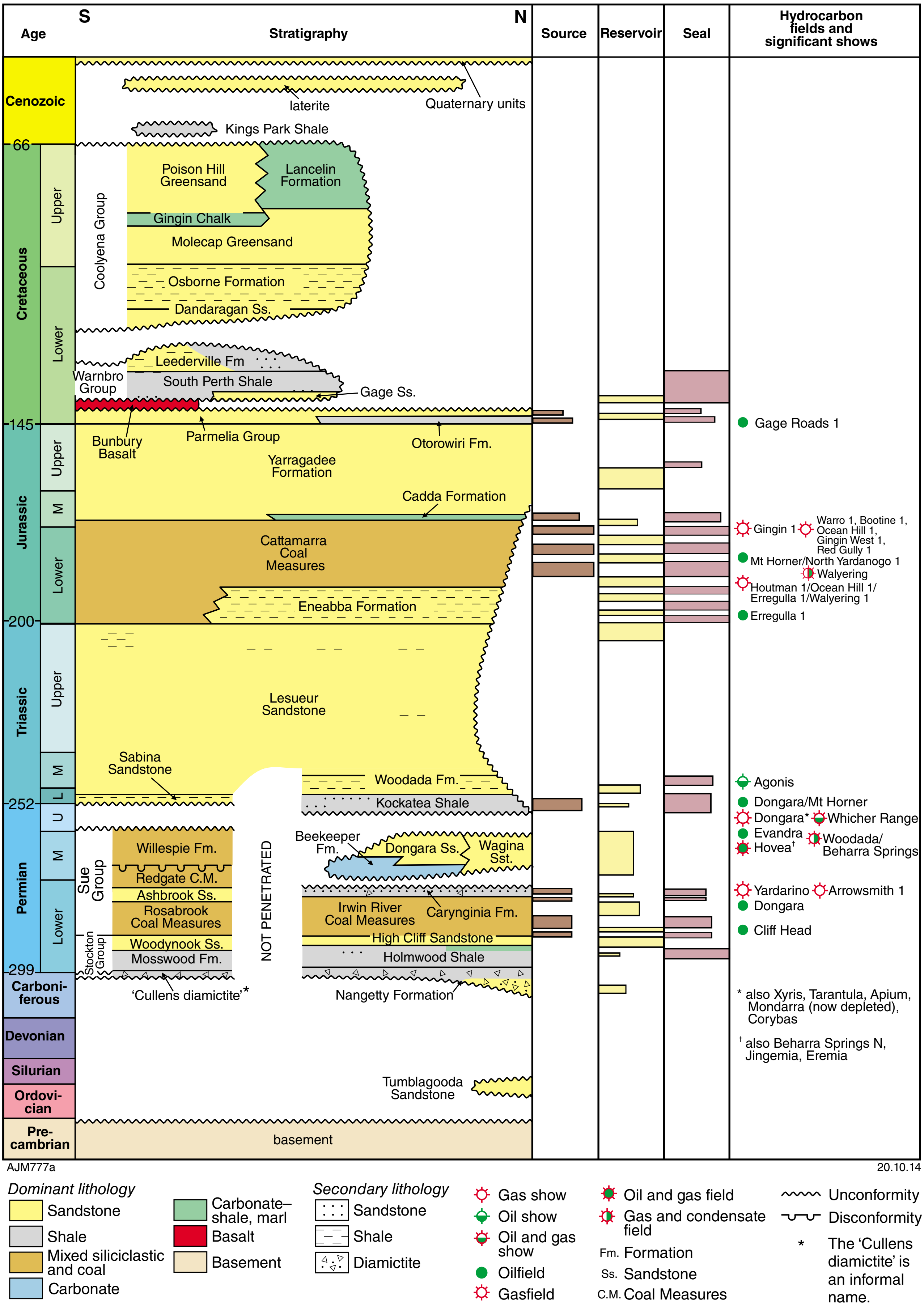

Figure 4. Generalized stratigraphic column for the Perth Basin

Table 1. Harvey 1 well data record

Well: GSWA Harvey 1			
Well classification	Stratigraphic	*Spud date*	00:00, 7 February 2012
TD[a] date	22:00, 8 March 2012	*Rig release date*	06:30, 26 March 2012
Final TD (m): MDRT/TVDSS	2945.0 / 2913.8	*Total rig days*	48
Completion status	Plugged and abandoned	*Permit/licence*	N/A
Surface location coordinates	Lat: 32°59' 30.730"S Long: 115°46' 28.093"E	*TD coordinates*	Lat: 32°59' 33.730"S Long: 115°46' 26.734"E
Surface location coordinates	385502.044E 6348947.564N	*TD coordinates*	385467.77E 6348860.13N
Reference	GDA94 / Zone 50 S	*Deviation at TD*	16.03°
Seismic location	80 m north of shot point 2450, 11GA-LL2	*Projection*	MGA94
Permanent datum	AHD	*RT to GL*	5.38 m
Ground Level (GL) to AHD	19.10 m	*RT to AHD*	24.48 m
Estimated final cost[b]	$9 085 000	*Native Title claim*	N/A
Location land owner	Alcoa	*Nature reserve*	No
Drilling contractor	MB Century	*Drilling rig*	Century 7
Drilling mud contractor	Rheochem	*Cementing services*	Halliburton
Mud logging services	Baker Hughes INTEQ	*Coring services*	Halliburton
MWD/LWD	Baker Hughes INTEQ	*Wireline logging*	Baker Hughes
Wellsite geologists	D Short and G Kjellgren		

Casing summary

	660 mm/26" hole	508 mm/20" casing	445 mm/17½" Hole	340 mm/13⅜" casing	311 mm/12¼" hole	244 mm/9⅝" casing	216 mm/8½" hole	178 mm/7" liner
MDRT (m)	-	-	53.7	53.7	845.6	841.5	2945	-
TVDRT (m)	-	-	53.7	53.7	845.5	841.4	2938.3	-
PIT[c]	-	-	-	N/A	-	2.46 SG EMW[d] at 848 m		-

Conventional coring summary

Core number	*Cored interval (m)*	*Formation/Member:*	*Bit/core size mm*	*Metres cored*	*Metres recovered*
1	895–931	Yalgorup Member	216/102	36	36.62
2	1266–1320	Yalgorup Member	216/102	54	53.16
3	1320–1336	Yalgorup Member	216/102	16	15.22
4	1336–1345	Yalgorup Member	216/102	9	7.76
2–4	1266–1345	Yalgorup Member	216/102	79	76.14
5	1896–1950	Wonnerup Member	216/102	54	51.64
6	2480–2534	Wonnerup Member	216/102	54	52.59

NOTES:

(a) TD = total depth

(b) Estimated final cost includes direct drilling costs and services, consumables, drilling camp, well site supervision, office support and HSE, insurances, site preparation and rehabilitation, geological and logging services and reporting, and laboratory costs.

(c) PIT = Pressure integrity test

(d) SG EMW = Specific gravity, equivalent mud weight

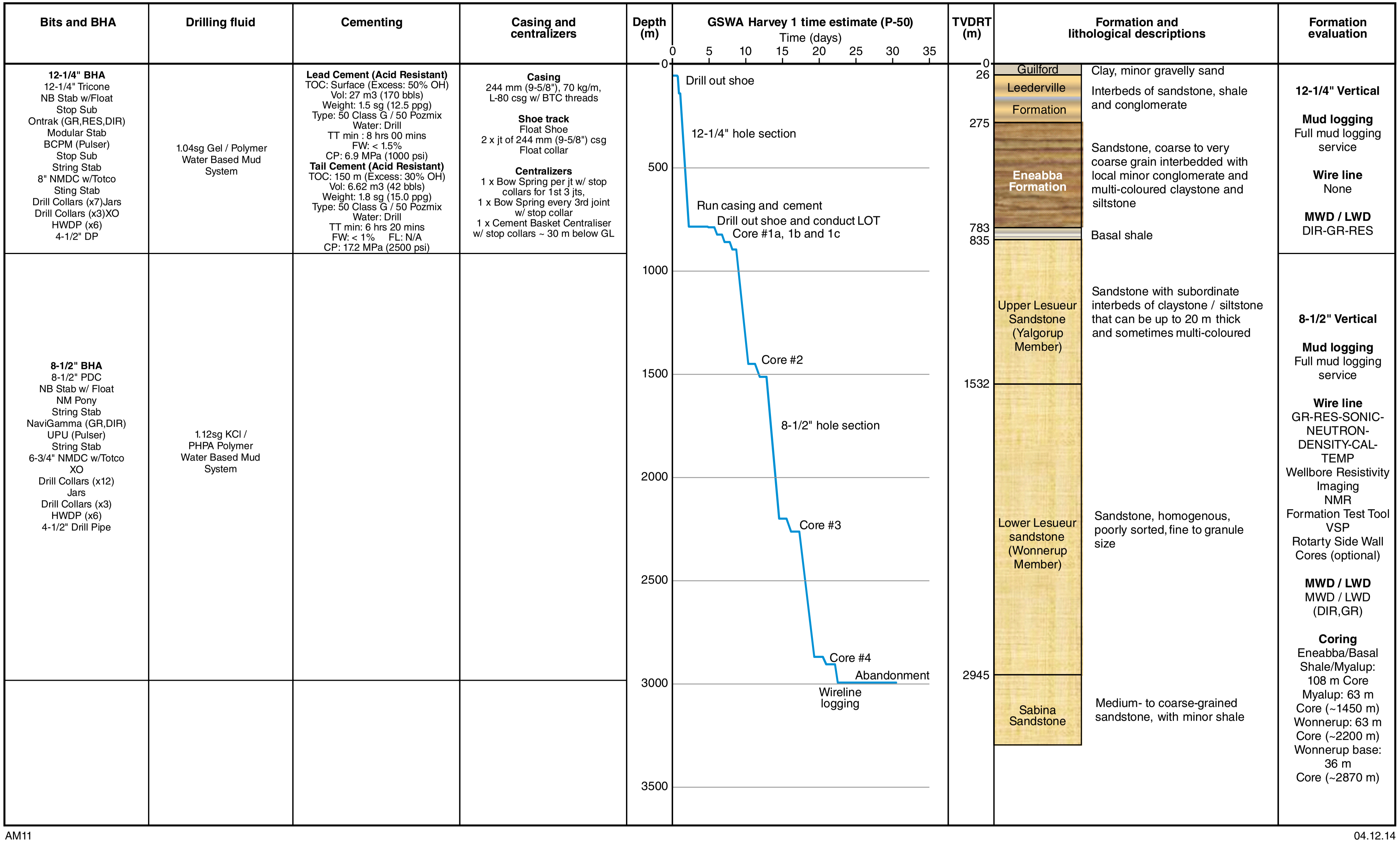

Figure 5. Harvey 1 well design and drilling program summary

Table 2. Prognosed versus actual depth to top of stratigraphic units in Harvey 1

Formation name	Prognosed depth to top of unit (m)			Actual depth to top of unit (m)				
	MDRT[(a)]	TVDSS[(b)]	Thickness	MDRT	TVDSS	High/low[(c)]	Difference	Thickness
Undifferentiated Quaternary	5	19	26	5.38	19.1	–	–	47.6
Leederville Formation	31	-7	244	53	-28.5	Low	-22	203.6
Eneabba Formation	275	-251	495	249	-224.5	High	26	375
Eneabba Formation, 'basal shale'[(d)]	770	-746	65	624	-599.5	High	146	76
Lesueur Sandstone	835	-811	2110	700	-675.5	High	135	2195
Yalgorup Member	835	-811	697	700	-675.5	High	135	678
Wonnerup Member	1532	-1508	1413	1378	-1353.5	High	154	1517
Sabina Sandstone	2945	-2921	50	2895	-2870.5	High	50	50
Dry hole, total depth	2995	-2971	–	2945	-2920.5	–	–	–

NOTES: (a) MDRT = measured depth rotary table
(b) TVDSS = true vertical depth subsea
(c) High/low = higher/lower than expected elevation
(d) The 'basal shale' is an informal unit of the Eneabba Formation

- natural gamma between 41 and 2882 m
- sonic (24") between 41 and 2871 m
- shallow and deep resistivity between 841.3 and 2886 m
- formation bulk density between 825 and 2864 m
- compensated neutron porosity between 796 and 2862 m
- spontaneous potential between 8 and 2596 m
- borehole temperature between 14 and 2855 m.

Alongside the wireline logs are displayed the cored intervals, biostratigraphy and vitrinite reflectance samples, a graphic mud log and a summary of abbreviated lithological descriptions directly from the mudlog.

The full wireline log suite is available to be downloaded from WAPIMS via DMP's web page and PDF versions of the logs are presented in Appendix 3. A graphic core log completed by the research group at Curtin University is included as Plate 2.

Daily reports

Daily reports for the rig move and drilling operations are contained in Appendix 2 and daily geological reports in Appendix 3.

Mud logging

A full mud logging program was completed by Baker Hughes INTEQ. Multiple sets of ditch cuttings were collected as outlined in the drilling and formation evaluation program (Appendix 1) and the mud logging report (Appendix 3). Cuttings were collected at 15 m intervals from 60 to 840 m, and 5 m intervals from 840 to 2945 m. An additional set of samples was collected for geochronology analysis by GA. The full mud logging and drilling report can be found in Appendix 3 along with lithological descriptions completed by the wellsite geologists.

Conventional coring operations

The original design of the program was to cut four core intervals to obtain representative material for geological, reservoir and seal testing. The first core was planned to sample an interpreted fine-grained unit at the base of the Eneabba Formation, as intersected in Lake Preston 1 and Pinjarra 1, and include a section of the upper Lesueur Sandstone below about 788 m. The second core was planned within the upper Lesueur Sandstone at about 1450 m, the third within the lower Lesueur Sandstone at about 2200 m and the fourth towards the base of the lower Lesueur Sandstone at about 2870 m.

The interpreted basal shale of the Eneabba Formation was intersected in the openhole section of the well, between about 625 and 704 m, and therefore not cored. The coring program was subsequently modified to obtain representative rock samples for analysis from the remainder of the well. Six cores covering four intervals were cut and recovered from Harvey 1; the depths are listed in Table 1.

Halliburton completed all coring operations. Once the core was pulled to the surface, a standard set of handling procedures was followed for each core:

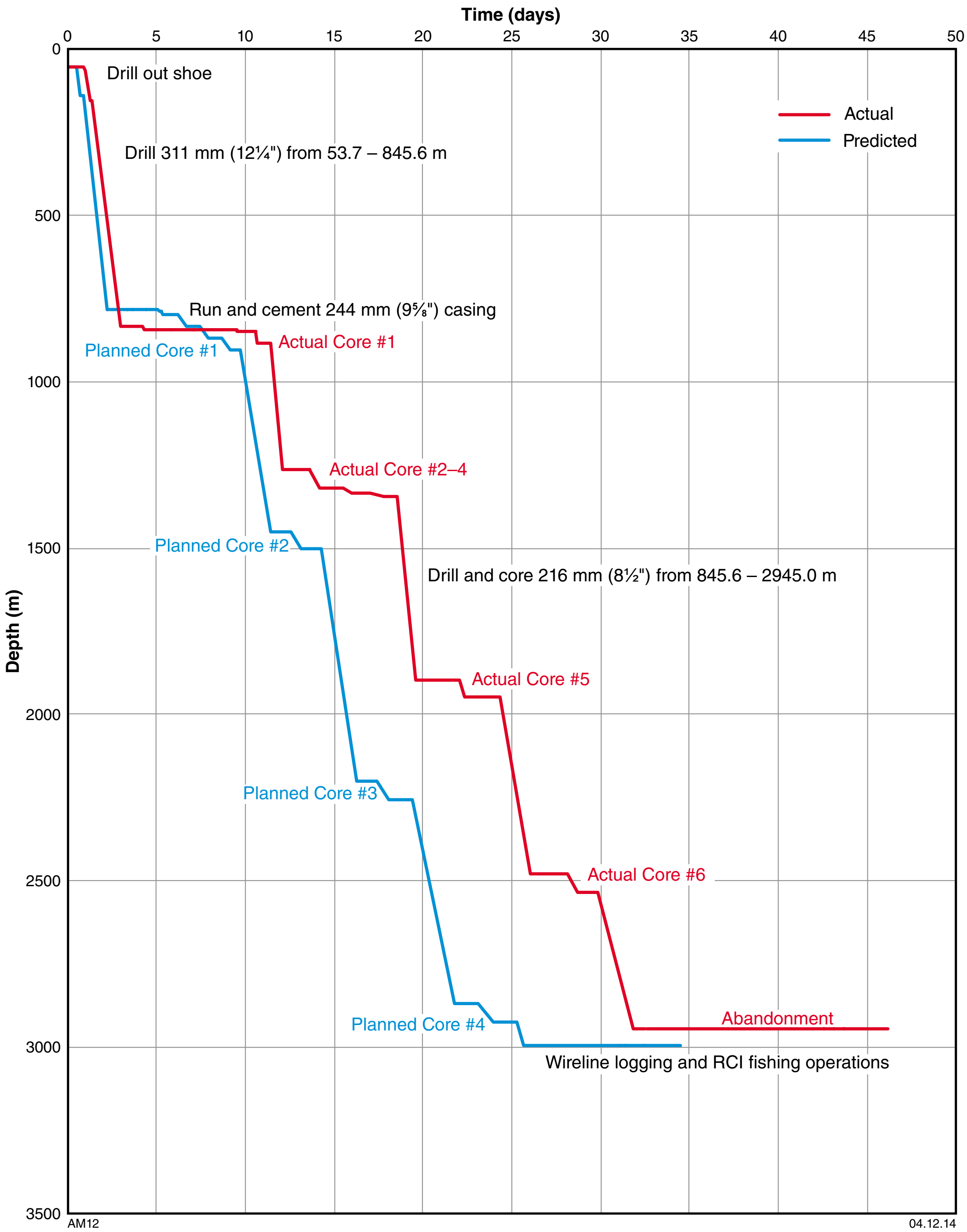

Figure 6. Harvey 1 drilling time curves: predicted versus actual

Table 3. Summary of porosity and permeability analyses for Harvey 1

Plug number and type[(a)]	*Depth (m)*	*Grain density (g/cc)*	*Porosity 800 (%)*[(b)]	*Permeability 800 (mD)*[(b)]	*Porosity 4300 (%)*[(b)]	*Permeability 4300 (mD)*[(b)]	*Comment*	*GSWA sample numbers*
1H	897.63	2.65	23.60	918.17	–	–	Friable sample	206601
2H	898.47	2.65	25.12	937.69	–	–	Friable sample	206602
3H	903.62	2.65	17.64	10.26	17.01	9.02	–	206603
4V	904.00	2.65	21.85	47.68	21.12	40.03	–	206604
5H	905.71	2.65	24.56	1 047.82	23.91	936.55	–	206605
6H	906.93	2.67	20.03	3.77	19.86	3.25	–	206606
7V	907.00	2.68	19.87	0.89	18.65	0.77	–	206607
8H	909.17	2.64	20.59	42.75	20.11	36.55	–	206608
9H	911.53	2.64	25.66	–	–	–	Friable sample	206609
10H	913.83	2.64	24.87	–	–	–	Friable sample	206610
12H	915.45	2.67	20.12	19.45	19.69	17.71	–	206612
13H	917.57	2.64	21.74	1 164.54	–	–	Friable sample	206613
14H	919.16	2.66	24.62	1 253.55	23.55	912.54	–	206614
15V	920.00	2.66	20.29	6.29	19.54	4.43	–	206615
16H	920.56	2.66	19.33	5.15	18.89	4.67	–	206616
17H	922.42	2.65	18.19	17.56	17.80	16.15	–	206617
18V	923.00	2.65	24.55	1 037.89	–	–	Friable sample	206618
19H	923.23	2.66	23.17	798.66	22.66	751.01	–	206619
20H	925.84	2.66	21.76	156.33	21.02	132.05	–	206620
21H	926.00	2.65	24.09	837.20	–	–	Friable sample	206621
22H	927.61	2.65	24.33	964.14	22.24	789.52	–	206622
23H	929.30	2.65	24.93	1 573.23	–	–	Friable sample	206623
24H	930.71	2.65	18.37	2.99	17.18	2.66	–	206624
25V	930.84	2.65	18.87	3.45	18.12	3.03	–	206625
26H	1 266.21	2.65	17.65	11.40	17.04	8.16	–	206626
27H	1 271.95	2.65	24.66	–	–	–	Friable sample	206627
28H	1 273.89	2.65	15.01	0.72	14.39	0.41	–	206628
34H	1 323.53	2.63	–	–	–	–	Irregular plug not suitable	206634
35H	1 323.93	2.65	18.50	12.40	17.77	8.63	–	206635
36V	1 324.00	2.65	18.34	25.83	17.04	17.52	–	206636
37H	1 325.83	2.66	18.37	38.19	17.40	34.50	–	206637
38H	1 329.94	2.75	6.70	< 0.01	6.57	< 0.01	–	206638
40H	1 331.05	2.66	13.95	6.62	12.87	4.95	–	206640
41H	1 331.55	2.65	18.37	77.19	17.78	66.32	–	206641
42H	1 337.41	2.68	10.25	0.07	9.93	< 0.01	–	206642
43H	1 342.60	2.65	–	–	–	–	Irregular plug not suitable	206643
44H	1 343.61	2.68	13.24	0.52	12.62	0.07	–	206644
45H	1 897.66	2.65	15.51	136.82	15.07	128.58	–	206645
46V	1 897.91	2.65	16.05	215.01	15.34	194.35	–	206646
47H	1 901.61	2.65	16.35	578.67	15.86	527.89	–	206647

Table 3. continued

Plug number and type[(a)]	Depth (m)	Grain density (g/cc)	Porosity 800 (%)[(b)]	Permeability 800 (mD)[(b)]	Porosity 4300 (%)[(b)]	Permeability 4300 (mD)[(b)]	Comment	GSWA sample numbers
48V	1902.92	2.65	13.92	7.28	13.37	5.89	–	206648
49H	1904.70	2.65	14.92	41.62	14.53	37.44	–	206649
50H	1908.71	2.64	13.56	7.77	13.17	6.75	–	206650
51V	1908.92	2.65	18.37	77.24	17.66	68.13	–	206651
52V	1915.00	2.65	14.84	5.81	14.18	4.55	–	206652
53H	1916.38	2.67	10.94	0.44	10.23	< 0.01	–	206653
54H	1919.30	2.65	15.54	24.37	15.35	21.86	–	206654
55H	1927.20	2.67	15.08	26.65	14.62	25.54	–	206655
56H	1929.40	2.66	14.52	100.39	14.11	93.60	–	206656
58V	1930.00	2.81	11.56	< 0.01	11.13	< 0.01	–	206658
59V	1931.00	2.67	16.07	170.12	15.67	156.12	–	206659
60H	1935.50	2.65	16.33	122.38	15.97	111.49	–	206660
61V	1940.00	2.72	14.42	1.27	13.65	0.13	–	206661
62H	1940.58	2.68	15.46	0.34	14.22	0.07	–	206662
63H	2480.66	2.67	12.90	9.76	12.37	8.73	–	206663
64V	2480.91	2.65	11.01	0.68	10.57	0.12	–	206664
65H	2485.53	2.65	12.36	29.98	11.96	26.92	–	206665
66V	2488.00	2.66	8.80	< 0.01	8.37	< 0.01	–	206666
67H	2488.14	2.69	5.24	< 0.01	5.04	< 0.01	–	206667
68H	2491.22	2.65	13.02	19.78	12.57	18.17	–	206668
69H	2491.56	2.65	13.67	343.66	13.13	322.90	–	206669
70H	2492.34	2.81	2.12	< 0.01	2.01	< 0.01	–	206670
71V	2495.93	2.66	11.33	0.43	10.58	0.01	–	206671
72H	2496.22	2.65	12.47	40.83	12.10	38.13	–	206672
73H	2501.29	2.65	12.51	22.37	12.16	20.14	–	206673
74V	2503.00	2.65	12.13	4.13	11.66	3.17	–	206674
75H	2503.46	2.65	13.56	231.44	13.30	216.08	–	206675
76H	2506.55	2.73	11.01	0.48	10.66	0.21	–	206676
77H	2508.29	2.65	14.96	33.67	14.47	30.37	–	206677
78H	2509.42	2.78	2.25	< 0.01	2.08	< 0.01	–	206678
79V	2510.00	2.66	11.87	1.65	11.65	1.29	–	206679
80H	2510.29	2.67	14.24	145.09	13.86	134.35	–	206680
81H	2513.34	2.67	11.42	3.59	11.02	3.03	–	206681
82H	2513.53	2.67	12.48	9.70	11.86	7.78	–	206682
83V	2516.00	2.67	13.83	60.40	13.59	55.73	–	206683
84V	2520.00	2.66	10.99	3.48	10.70	2.37	–	206684
85H	2520.44	2.65	13.87	115.64	13.63	107.99	–	206685
86H	2523.34	2.71	2.36	< 0.01	2.08	< 0.01	–	206686
87B V	2525.00	2.65	12.23	3.79	11.78	2.97	–	206687
88H	2525.83	2.65	12.33	5.32	11.86	4.68	–	206688
89H	2527.38	2.65	11.94	9.44	11.49	7.51	–	206689
90H	2529.29	2.65	13.98	67.70	13.30	61.64	–	206690

NOTES: (a) H = horizontal plug, V = vertical plug
(b) Samples without values were not suitable for determining porosity and permeability

Table 4. Summary result of MICP tests for Harvey 1

Sample number[a]	Depth (m)	Corrected Mercury porosity (%)	Mercury/air displacement pressure (psi)	Pore throat distribution			Swanson method	
				Micropores < 0.1 micron (% PV)[b]	Intermediate 0.1-3 micron (% PV)	Macropores > 3 micron (% PV)	Brine permeability (mD)	Air permeability (mD)
1H	897.63	22.613	1.993	13.623	37.348	49.030	554.154	580.881
3H	903.62	20.280	20.300	26.192	69.260	4.547	2.352	5.799
9H	911.53	24.414	1.364	13.901	33.007	53.091	762.102	759.971
16H	920.56	18.440	51.014	40.431	59.569	0.000	0.391	1.277
22H	927.61	24.927	2.192	14.290	38.202	47.508	313.794	359.568
26H	1266.21	15.657	45.287	23.132	76.868	0.000	2.755	6.627
27H	1271.95	21.097	3.880	6.915	25.043	68.042	229.585	276.269
28H	1273.89	16.023	43.234	31.090	68.910	0.000	0.466	1.482
35H	1323.93	16.434	10.248	2.273	78.337	19.390	5.882	12.565
36H	1324.00	20.431	10.245	10.538	45.466	43.997	33.794	54.897
38H	1329.94	13.163	92.283	51.886	48.114	0.000	0.098	0.396
42H	1337.41	14.230	50.107	46.052	53.948	0.000	0.178	0.657
43H	1342.60	11.693	44.358	62.858	37.142	0.000	0.080	0.336
44H	1343.61	16.475	45.122	27.873	72.127	0.000	0.762	2.241
45H	1897.66	13.618	11.294	17.199	50.684	32.117	8.982	17.955
46H	1897.91	13.279	1.129	13.772	43.204	43.023	50.650	77.225
47H	1901.61	16.733	2.916	9.205	39.426	51.370	172.229	216.793
48H	1902.92	2.206	41.385	31.992	68.008	0.000	0.011	0.063
53H	1916.38	9.965	84.829	40.249	59.751	0.000	0.056	0.249
60H	1935.50	16.383	5.717	12.056	54.123	33.821	15.136	27.883
62H	1940.58	11.725	62.203	36.559	63.441	0.000	0.150	0.568
63H	2480.66	11.230	41.279	21.334	78.666	0.000	0.367	1.210
64H	2480.91	10.384	14.699	24.734	68.521	6.745	0.380	1.245
69H	2491.56	12.536	5.190	11.139	48.126	40.735	29.068	48.347
72H	2496.22	11.815	6.943	26.754	61.608	11.638	0.897	2.571
75H	2503.46	12.336	3.881	8.682	38.776	52.542	59.481	88.436
83H	2516.00	14.210	6.301	15.068	35.483	49.448	32.197	52.700
88H	2525.83	9.885	49.901	31.745	68.255	0.000	0.216	0.773

NOTES: (a) H = horizontal plug
(b) PV = pore volume

The upper Lesueur Sandstone sample (GA 2132432), at 1000 m depth, yields dates between 3262 and 511 Ma. Three age components have been identified; the oldest is Mesoproterozoic (1272–1067 Ma), the second is Mesoproterozoic–Neoproterozoic (1011–866 Ma), and the younger cluster is Neoproterozoic (604–511 Ma). Analyses yielding dates >1300 Ma are dispersed and do not form significant age components.

The Eneabba Formation sample (GA 2132431), at 450 m depth, yields dates between 2681and 515 Ma. The data define two age components; the older is Archean (2681–2636 Ma) and the younger is Mesoproterozoic (1223–1080 Ma). The remaining results are dispersed (between 2636 and 1223 Ma, and <1080 Ma) and do not form significant age components.

The shallowest sample analysed was from the Leederville Formation (GA 2132430), at 100 m depth. Data are dispersed between 3358 Ma and134 Ma, with two dominant and, arguably, two subordinate age components. The oldest dominant component is Archean (2763–2571 Ma) and the younger dominant component is Mesoproterozoic (1232–1061 Ma). The older subordinate component is Paleoproterozoic–Mesoproterozoic (1763–1595 Ma) and the younger is Neoproterozoic (727–683 Ma). The remaining results are dispersed and do not form significant age components.

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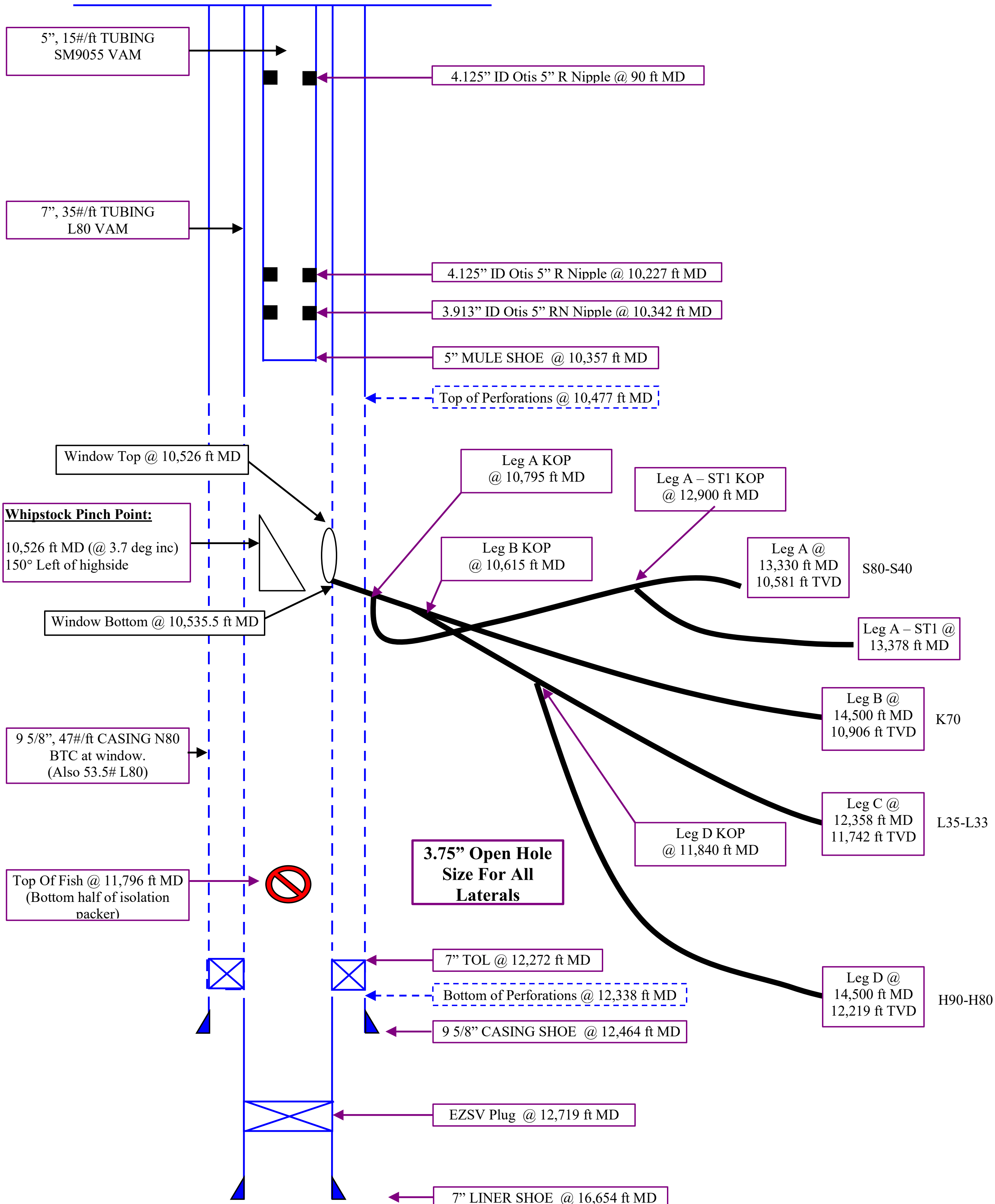
Wellbore Diagram



Chronological Overview

Past Well Work

This well was spudded on May 10, 1980 and drilled to a total depth of 16,656 ft KB. The Thamama was topped at 10,470 ft KB and 9 5/8 inch casing set at 12,464 ft KB with a 7 inch liner being set from 12,272 ft KB to 16,464 ft KB. During initial drilling of the well, tests were conducted through 7 inch liner in the lower and middle parts of the Jurassic. Both these tests produced dry sour gas with appreciable amounts of water (115-200 bbls/mmf). Another test was conducted in the upper part of the Jurassic, which produced gas and condensate (about 100 bbls/mmcf of gas) with only a trace of water and small amount of H_2S in the gas (0-50OPPM). The estimated net pay in the well is 100 in the upper part of the Jurassic and 825 net feet of pay in the Thamama. Four tests were conducted in the Thamama, all of which produced various amounts of gas and condensate (about 100 bbls/mmcf) and a trace of water. All of the Thamama zones that were tested as well as the uppermost zone tested in the Jurassic were squeezed with cement and the well was suspended. The well was re-entered, and completed with 146 ft of perforations in the upper part of the Jurassic and 174 ft of perforations in the lower part of the Thamama. Perforation shot density was 4 SPF. There is approximately 651 net feet of Thamama pay up the hole, which has not been perforated. Cement bond logs show marginal to no cement over the perforated areas.

During re-entry of the well, several sets of perforations were individually acidized, flow tested for a short period then killed with calcium carbonate pills, and an EZSV bridge plug set above them. The acid jobs, which were performed on individual sets of perforations, consisted of 200 gallons of 28% acid per foot of perforations and utilized diverter plugs consisting of 500 gallons of gel, rock salt and Halliburton TLC-8O chemicals. Prior to final completion, all bridge plugs were drilled out and a permanent 5 1/2 inch completion string with packer was run. After running final completion string, a short flow test of all zones combined, indicated a lower rate than was expected based on summing up individual short-term tests. Production logs were run in the well and these indicated that only about 54% of the zones were contributing to production and that no production was being contributed by the upper Jurassic perforations.

A large acid job consisting of 400 gallons of 15% acid per foot of (128,000 gallons or 100 gals/ft) and -1920 -- 1.0 sp gr (50% excess over holes in pipe) at 50 BPM was then performed on a II open zones in the weII. This acid job, based on short tests, increased productivity from 48.9 mmcfd with FTP of 3189 psig on a 1 inch choke to 61.9 mmcfd with a FTP of 4836 psi on a 1 inch choke. Production logs run after this acid job showed very little change in flow distribution with still no production being contributed by the upper Jurassic perforations. It is noted that production logging tools could not be run below 13,180 ft KB (top of upper Jurassic perforations) as the tool seemed to meet an obstruction at this depth. There were no indications of problems at this depth while drilling or completing the well. The well was placed on production in July 1982 and produced at the rate of nearly 90 mmcfd on a 1 3/8 inch choke 3700 psi FTP.

During the fishing operations, shortcomings of the snubbing unit became evident. The snubbing unit weight indicator was insensitive due to the buoyancy of the pipe and could not put a significant amount of weight on mills, fishing tools, etc. A normal rig may have been able to fish successfully. The snubbing unit did allow the well to be produced while running in the hole with fishing tools. The well was produced 26 days and/or nights during the repair.

Intervention History (Sajaa 1):

11.Sept.2001	Wellhead dropped by 6". This was at the end of work to change flowline spool flange closest to tree which had been found to be very badly eroded/corroded.
24.Sept.2001	Cellar dug out, clamp installed between 30" and 20" and concreted in place.
20.Nov.2001	Made up and ran 3.60" gauge cutter. Found obstruction @11504ftbrt.
21.Nov.2001	Made up and ran 2.00" gauge cutter. Found obstruction @11504ftbrt. Ran tandem Amerada gauges with 10-minute stops @5000/8000/10000 &11450ft (wl). Confirmed data captured.
Nov 2003	Surface PBU built up to 685 psi after 7 day shut in
19.Jan.2004	Ran 3.80" gauge cutter. Jarred through obstruction @ 11,187' rkb and could not past 11,476' rkb. (HUD @ 11,456' WLM 11,476')
26.Jan.04	PMIT caliper survey from 11,025 ft to 9,968 ft. Indication of scale at top perfs.
04.Feb.04	Repeat PMIT caliper survey 10,590 – 10,115 ft. Confirmation of scale at top perfs.
08.Feb.04	Ran 3.845" gauge cutter to 11,471 ft rkb. Carried out static pressure survey.

foam sweep was pumped to clear out the wellbore and POOH was commenced after the sweep was seen to round the bit. Water rate was reduced to 8 gpm, N2 rate to 1,000 scfm after bottoms up time (10 minutes) to minimize vibration on the BHA. The BHA was pulled to 10,660 ft and a gamma ray survey was conducted (avoiding passing the BHA through the window). The BHA was then picked up and the Leg B KOP trough was time drilled between 10,615 ft – 10,630 ft.

A one hour flow test was conducted on a constant 90/64th inch choke with the BHA in the 7" casing section above the window (at 10,500 ft). Results of the flow test are given below:

Results of Leg D flow test with BHA at 10,500 ft and choke at 90/64ths:

Time	7" FCSGP [Psig]	FWHP [Psig]	BHP [Psia]	Qgas [MMscfd]	Q Condensate [BPD]	Q Water [BPD]	Solids [%]
16:15	599	134	778	3.50	0	678	0
16:30	590	132	768	3.86	0	681	0
16:45	593	160	771	3.47	0	667	0
17:00	598	125	776	3.19	0	586	0
17:15	600	155	777	3.09	0	615	0

The BHA was pulled out of hole and undeployed prior to conducting bi-weekly BOP pressure testing.

Drilling the Third leg - Leg B

BHA #10 was RIH to drill the build section, with a 2.9 deg AKO, and passed through the window at the first attempt. BHP at the window was 1,060 psi with a 7" casing pressure of 821 psi.

Once drilling continued the initial FBHP was at 970 psi with a 7" casing pressure of 740 psi. FBHP was reduced to 940 psi before returns were relatively stable. As we drilled on the FBHP was falling as the charging caused by closing in the well during the BOP testing reduced. Once at 10,770 ft the choke was at a 74 / 64th, and the FBHP was still dropping along with returns. As drilling continued the returns were falling faster than FBHP was i.e. the well was starting to approach balance. The end of the build section was reached at 10,815 ft and the well was shut in on the trip out of the hole.

BHA #11, a kick off assembly comprising 3.75" M09 bit, 2 7/8" M1 ADM motor with 0.7 degree AKO was picked up and RIH. The BHA was oriented to 150 deg Left and passed through window. There was an electrical short in the BHA at 10,800 ft, shortly after opening up well to the BRT. BHA #11 was POOH and undeployed.

BHA #12, a kick off assembly comprising 3.75" M09 bit, 2 7/8" M1 ADM motor with 0.7 degree AKO was picked up and RIH. The BHA passed through the window with no

NPT Analysis

NPT for Sajaa 1 was 13.1% of total time, 11.5% of total cost. The information is broken down in the tables & charts below:

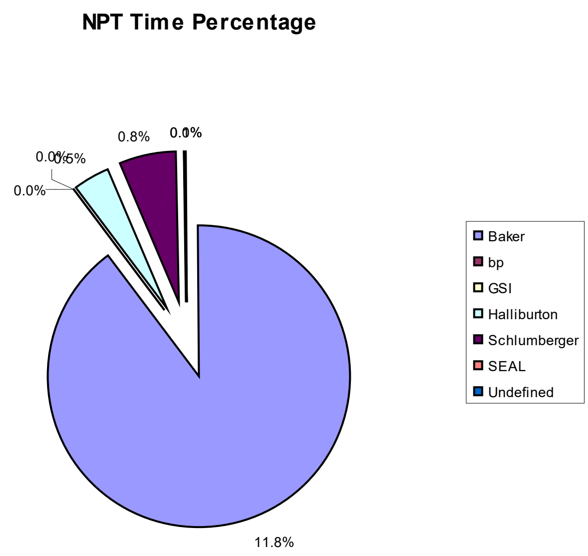
Totals	Time [Hours]	Cost	% of Total Time	% of Total Cost
Baker	79.0	$355,106	11.8%	9.8%
bp	0.0	$0	0.0%	0.0%
GSI	0.0	$0	0.0%	0.0%
Halliburton	3.5	$20,778	0.5%	0.6%
Schlumberger	5.2	$41,209	0.8%	1.1%
SEAL	0.0	$0	0.0%	0.0%
Undefined	0.4	$1,413	0.1%	0.0%
Total NPT	88.1	$418,506	13.1%	11.5%
Total Well	672.0	$3,631,750		
Productive	583.9	$3,213,244	86.9%	88.5%

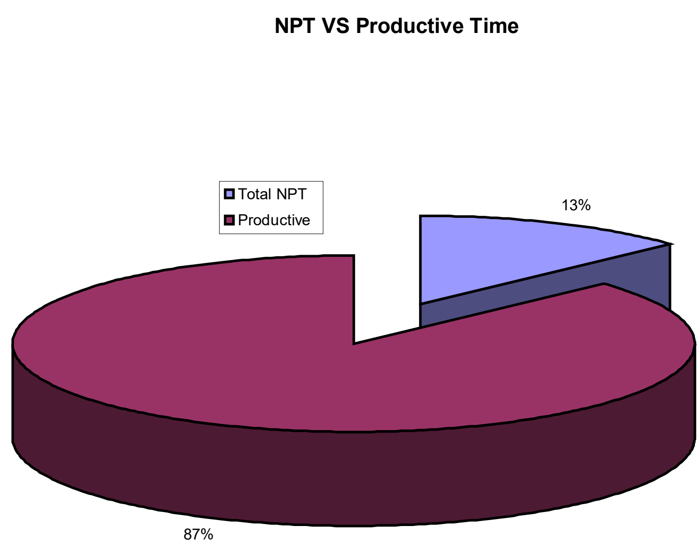
Rig Days 28.0Days
Total Well Cost $3,631,750

Every failure of a BHA was recorded as NPT against Baker (NB: this includes BHI & BOT) as per bp standard procedure, to remove any subjectivity from the appliance of NPT criteria. However, bp considers the use of 2 milling BHA's per window (1 to run Whipstock and one to mill the window) and 2 drilling BHA's per leg as a reasonable technical limit for this project, given the downhole temperatures and drilling fluid pumped. Accordingly, the NPT for BHI using these criteria would be reduced to:

Totals	Time [Hours]	Cost	% of Total Time	% of Total Cost
Baker	**21.1**	**$94,695**	3.1%	2.6%
bp	0.0	$0	0.0%	0.0%
GSI	0.0	$0	0.0%	0.0%
Halliburton	3.5	$20,778	0.5%	0.6%
Schlumberger	5.2	$41,209	0.8%	1.1%
SEAL	0.0	$0	0.0%	0.0%
Undefined	0.4	$1,413	0.1%	0.0%
Total NPT	30.2	$158,095	4.5%	4.4%
Total Well	672.0	$3,631,750		
Productive	641.8	$3,473,655	95.5%	95.6%

Rig Days 28.0 days
Total Well Cost $3,631,750


NPT Time Percentage
0.0%
0.5%
0.8%
0.0%
0.1%
0.0%
11.8%
Baker
bp
GSI
Halliburton
Schlumberger
SEAL
Undefined


NPT VS Productive Time
Total NPT
Productive
13%
87%

NPT Time Breakdown:

BAKER				
Date	**Time**	**Problem**	**Daily Cost**	**NPT cost**
14th Feb	14.33	Short in Whipstock BHA#2	$95,255	$56,875
16-Feb	8.67	Mill on BHA #5 worn out - replace	$96,214	$34,757
17-Feb	1.00	Mill on BHA #5 worn out - replace (contd.)	$101,787	$4,241
21-Feb	12.10	BHA communications problem (BHA # 8)	$124,502	$62,770
28-Feb	10.43	Short in BHA #11	$101,969	$44,314
28-Feb	6.60	Orienter Failure in BHA #12	$101,969	$28,041
29-Feb	4.37	Orienter Failure in BHA #12 (contd.)	$101,969	$18,567
29-Feb	7.13	Short in BHA # 13	$109,011	$32,385
29-Feb	3.47	Short in BHA # 13 (contd.)	$109,011	$15,761
29-Feb	0.95	Orienter Failure in BHA #14	$109,011	$4,315
1-Mar	9.97	Orienter Failure in BHA #14 (contd.)	$127,773	$53,079
Totals	**79.02**		**Dollars**	**$355,106**

BP				
Date	Time	Problem	Daily Cost	NPT cost
				$0
Totals	**0.00**		**Dollars**	**$0**
BP				

SLB				
Date	**Time**	**Problem**	**Daily Cost**	**NPT cost**
13-Feb	1.58	Adjust faulty gooseneck support arm (coil in hole)	$245,805	$16,182
13-Feb	1.50	Replace gooseneck support arm (coil out of hole)	$245,805	$15,363
27-Feb	1.13	Leaking new N2 flow meter - no gaskets fitted on Graylok flanges	$105,882	$4,985
29-Feb	1.03	Flush Power pack radiator with water to unblock it.	$109,011	$4,678
Totals	**5.24**		**Dollars**	**$41,209**

GSI				
Date	**Time**	**Problem**	**Daily Cost**	**NPT cost**
			n/a	$0
			n/a	$0
				$0

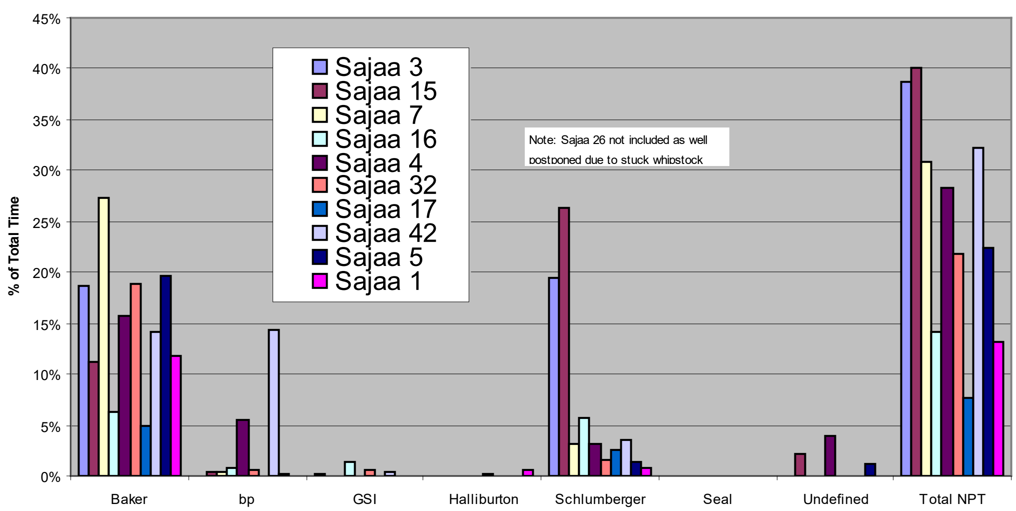
HES				
Date	**Time**	**Problem**	**Daily Cost**	**NPT cost**
24-Feb	3.28	Make long wiper & tie in new condensate pump	$141,268	$19,307
24-Feb	0.25	Condensate caused BHI computer to be reset	$141,268	$1,472
Totals	**3.53**		**Dollars**	**$20,778**

Misc.

Date	Time	Problem	Daily Cost	NPT cost
18-Feb	0.35	Problem with crane computer system while undeploying.	$96,897	$1,413
Totals	0.35		Dollars	$1,413
Misc.				

Seal				
Date	Time	Problem	Daily Cost	NPT cost
				$0
Totals	0.00		Dollars	$0
Seal				

Service Company NPT Comparison For Sajaa 3, 15, 7, 16, 4, 32, 17, 42, 5 & 1



Appendices

Appendix 1: HSE Statistics

Environmental Totals

	Gas thru Export Line (MMscf)	Gas flared (MMscf)	Cold Vent (MMscf)	Condensate To Plant [bbl]	Water To Land Farm (bbl)	Average ppm	Water Pumped to Formation (bbl)
Well Totals	31.074	31.534	18.36	270.0	9,758.3	4,606	8,228.6

37,408 man hours on this well without a DAFWC.
394,832 man hours so far on the project without a DAFWC.

Appendix 9: CT Specifications

A tapered work string with no e-line was used for BHA's # 1 (under-reaming) and BHA's 4-6 (window milling). A tapered e-line string was used for BHA's 2-3 (setting whipstock) and BHA's 7-15 (drilling). This was done to maximize the life of the tapered e-line string. The details of both strings are listed below:

String No. PE 10586 (Tapered Work String)

Coiled Tubing String Segments

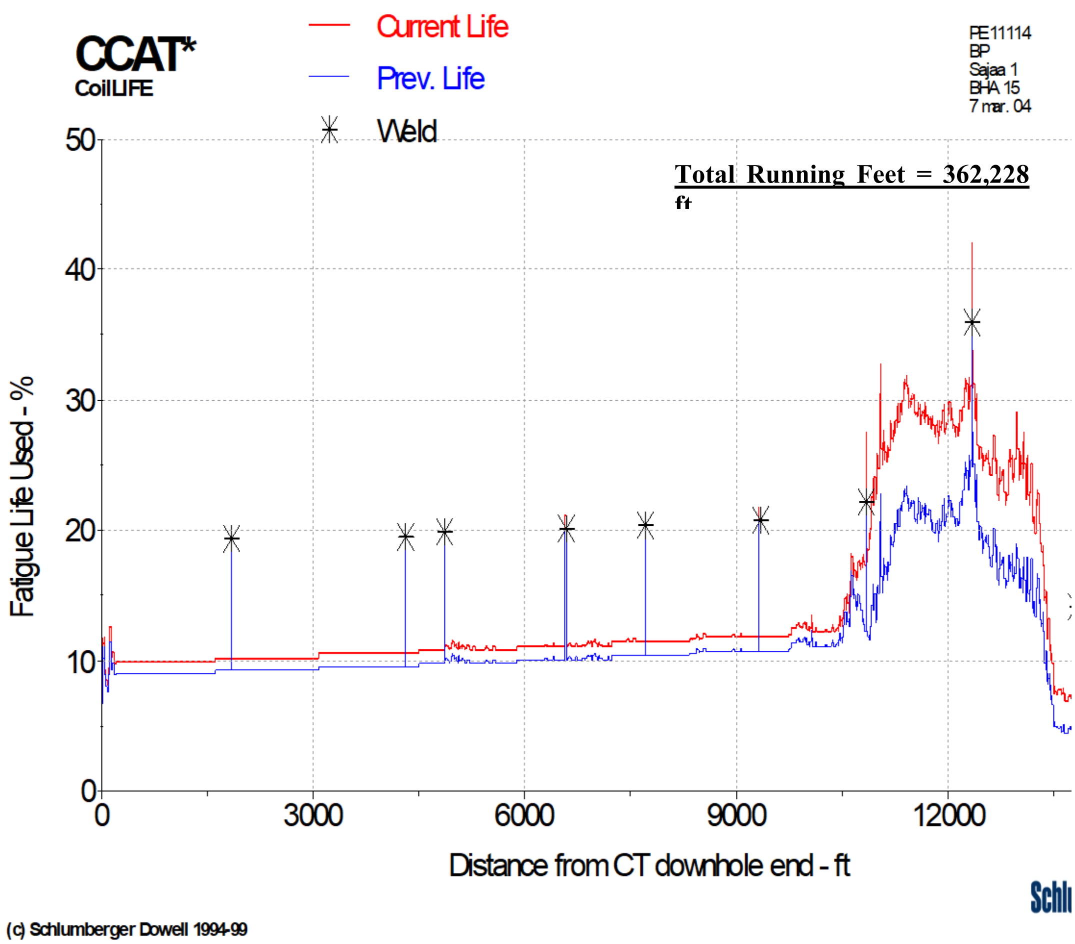
Reel Identification	**work reel**
Coiled Tubing Material	HS-90
Reel Core Diameter	98.000 in
Reel Core Width	96.000 in
String Identification	**pe 10586-8jan04**
Total String Length	12345.00 ft
CT Total Volume	51.0 bbl

Coiled Tubing String Segments

Outer Diameter (in)	Nominal Thickness (in)	Minimum Thickness (in)	Segment Length (ft)	Segment Volume (bbl)
2.375	0.134	0.128	5065.00	21.8
2.375	0.145	0.138	1415.00	6.0
2.375	0.156	0.148	1225.00	5.1
2.375	0.175	0.167	1610.00	6.4
2.375	0.190	0.183	3030.00	11.7

Job #	Job Date	Job ID	Company	Well Name	Job Description	GN Radius (in)	H_2S Level (ppm)
1	11 Feb 04	BHA 1	BP	Sajaa 1	Scale Milling	100.00	80
2	15 Feb 04	BHA 4	BP	Sajaa 1	Window Milling	100.00	80
3	17 Feb 04	BHA 5	BP	Sajaa 1	Window Milling	100.000	80
4	18 Feb 04	BHA 6	BP	Sajaa 1	Window Milling	100.00	80

Coiled Tubing Fatigue Plot


CCAT*
CoilLIFE
Current Life
Prev. Life
Weld
FE11114
BP
Sajaa 1
BHA 15
7 mar. 04
Total Running Feet = 362,228 ft
Fatigue Life Used - %
50
40
30
20
10
0
0
3000
6000
9000
12000
Distance from CT downhole end - ft
(c) Schlumberger Dowell 1994-99

Society of Petroleum Engineers

SPE Technical Report

Getting to Zero and Beyond: The Path Forward

Improving Safety in the Oil and Gas Industry

March 2018

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The authors wish to thank all the contributors and peer reviewers for their time, effort, and contribution to this report.

This report represents the consensus viewpoint of subject matter experts and is intended to provide useful guidance to SPE members, the industry, and public.

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2.0 Introduction

For me, the idea is to shun the incremental and go for the leap.
–Jack Welch, Chief Executive Officer of General Electric, 1973–2001

The oil and gas industry was started by visionaries and wildcatters who foresaw a brighter future and were willing to scrap the status quo and take steps to achieve their vision. For the industry to reposition itself for the future today, it needs an injection of the same willingness to cast aside some strongly held legacy beliefs and practices and take the next leap forward—achieving zero harm.

Rod Gutierrez, PhD, Global Leader—Culture and Change Management for DuPont Sustainable Solutions, has written that the concept of zero has encouraged industries to move away from simply minimizing risks through engineering out hazards toward actively engaging people for an overall improvement to workplace health and safety outcomes (Gutierrez 2015):

> *But however positive this advancement, it is important to consider what comes next. What if we are to shift our point of reference, so that zero harm was no longer the goal, but rather simply the beginning of sustained workplace health and safety? By reconfiguring the scale that we apply to workplace health and safety, by no longer considering zero as a final objective but rather a starting point, a minimum standard, we allow for the opportunity to initiate comprehensive and meaningful outcomes*.

The concept of zero harm is not a new one in the oil and gas industry, which has for more than two decades set HSE goals that are focused on the reduction of incidents with the ultimate intended outcome that no one be hurt and no releases occur. However, the challenge for decades has been the alignment of industry on an effective pathway to achieve these goals. To attain zero harm, a step change in thinking, performance, and alignment around HSE is required across the industry. 2016 SPE President Nathan Meehan stated, "This trend in performance improvement over the past decade has plateaued. We need a breakthrough" (Meehan 2015).

It is acknowledged that some believe directing attention toward achieving zero harm can lead to a detrimental focus on incidents and injuries that may lead to underreporting of incidents, gaming of statistics, manipulation of incident definitions, and overly aggressive injury case management. Sidney Dekker has argued against zero being a goal. However, his explanation of it as a vision perhaps best illustrates what it can represent (Dekker 2013):

> *A zero vision is a commitment. It is a modernist commitment, inspired by Enlightenment thinking, that is driven by the moral appeal of not wanting to do harm and making the world a better place. It is also driven by the modernist belief that progress is always possible, that we can continually improve, always make things better*.

Between 2009 and 2016, SPE facilitated a series of global sessions to develop ideas for the continued improvement of HSE in the industry. These sessions brought together leaders representing diverse disciplines from across the industry, government, and academia to discuss a simple question: How can the oil and gas industry achieve zero harm?

The participants of the SPE sessions identified that achieving and sustaining zero harm relies on the industry's people and the factors that influence the interaction of people with each other, with the facilities

Workshop participants developed and endorsed "Getting to Zero" vision and mission statements **(Fig. 1)** that introduced a human-factors approach to safety and sought to be both understandable and inspiring.

> ***"Getting to ZERO"***
> **A Vision Statement**
> As a member of SPE, "Safety **IS** my way of Life" at work, home, and in my community. Safety is fully integrated into work knowledge, skills, and abilities, regardless of my position or responsibility in the organization.
>
> As a core value, I am accountable for my safety and the safety of my coworkers, family, and friends. Indeed, I believe every employee associated with the Petroleum Industry is equally accountable for their own safety as well as the safety of those with whom they work and live.
>
> I passionately believe ZERO injuries/incidents is achievable because "SAFETY **IS** MY WAY OF LIFE."
>
> **A Mission Statement**
> The success of our business depends on my Safety Ethic to properly plan my work; to ensure my tools are available, maintained, and safely utilized; to know the hazards of my work and control or eliminate exposure to them; to follow all procedures safely; to *"STOP WORK"* when I deem it is necessary; to immediately report all near-misses and accidents; and to make safety improvement suggestions when warranted.

Fig. 1—SPE Getting to Zero session participants' vision and mission statements, generated as a work product from 2010 SPE Forum Series Getting to Zero—An Incident-Free Workplace: How Do We Get There?

Following these two events, the SPE Health, Safety, Security, Environment, and Social Responsibility (HSSE-SR) Advisory Committee accepted responsibility for further broadening the discussion. In preparation for the 2016 SPE International Conference and Exhibition on HSSE-SR in Stavanger, a series of interactive sessions was conducted. These sessions began in 2015 and were held across the globe (including North America, South America, Europe, Asia, and the Middle East), with the last session organized in Stavanger in conjunction with the April 2016 SPE conference.

More than 750 participants from numerous disciplines and organizations joined the global sessions. (See Appendix 7.2 for information on the participants' organizational affiliations and employment demographics.) The sessions integrated both in-person and live-webinar formats, encouraging active participation including question-and-answer sessions and real-time polling of participants. The discussions focused on three questions:

- Is getting to zero achievable?
- What are the most critical (organizational) values to achieving zero?
- What issues to achieving zero exist that require additional effort and time to understand and resolve?

Nearly 70% of the participants agreed that achieving an expectation of zero harm was possible for the industry, with approximately 20% recognizing it may be possible. Additionally, participants identified that the current HSE culture and risk tolerance across the industry were likely the most significant factors in positioning the industry to regard zero as an achievable expectation.

Participants prioritized five organizational values that define culture and impact the ability of the industry to achieve an expectation of zero harm **(Fig. 2).**

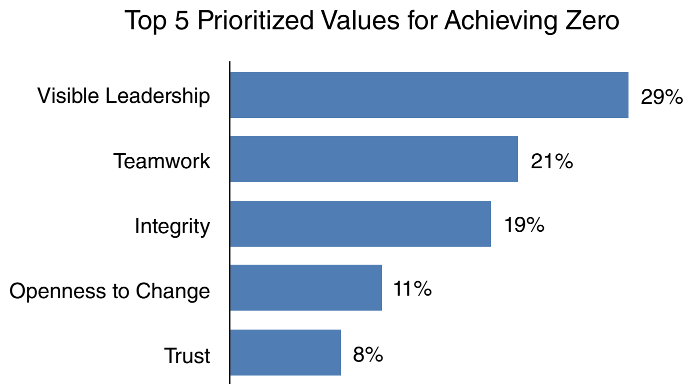

Fig. 2—SPE Getting to Zero sessions participant responses to survey question: Select five values that you believe are critical to achieving a safety vision of "getting to zero."

Participants also identified five obstacles (listed in priority as defined by the participants) that require attention in the near term if the industry is to achieve an expectation of zero harm:

- Insufficient alignment and application of human factors
- Misalignment and confusion on the expectation of zero harm
- Ineffective leadership development
- Inaccurate identification and management of risk
- Market pressure and recent industry downturn

Participants overwhelmingly agreed that the evolution of the industry's HSE culture and risk-tolerance maturities were driving forces in the recognition that human factors are pivotal and a required pathway for the industry to achieve an expectation of zero harm.

If the oil and gas industry is to achieve the needed step change in culture, risk management, and performance to embrace and realize an expectation of zero harm, it is worthwhile to understand more about human-factor elements **(Fig. 3).** Specifically, how they may be and are being applied across the oil and gas industry and how HROs and other industries changed, learned, and worked together to fulfill the expectation of zero harm in their operations.

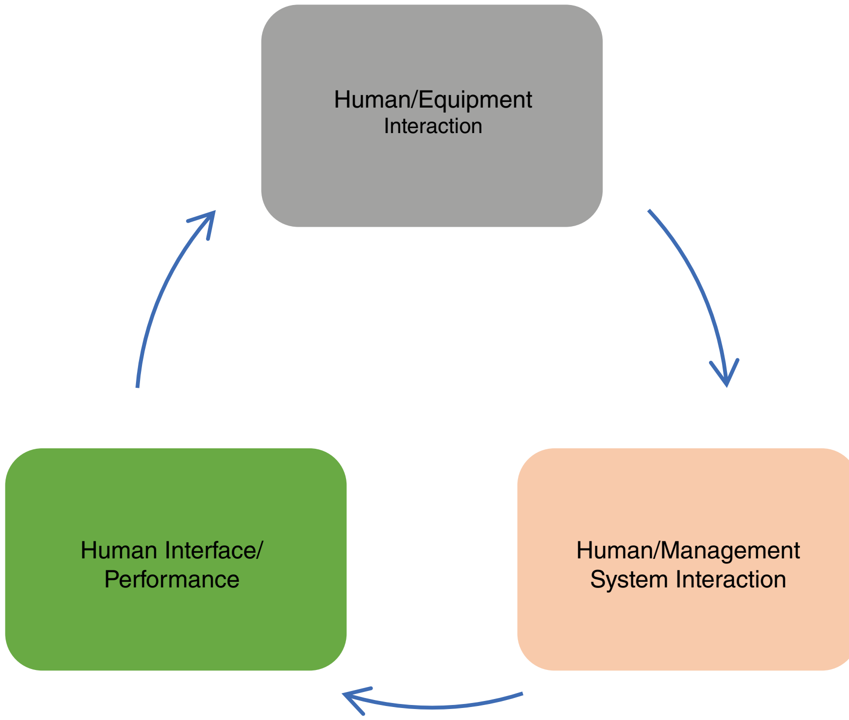

Fig. 3—Human-factors elements.

4.1 Human/Equipment Interaction

Addressing human/equipment interaction requires mastering three categories:

- Identification of human actions that are critical to safety
- Review of legacy-equipment incident data with proper human-factors causal analysis to identify opportunities to improve equipment design
- Incorporation of error-reducing strategies and questions during risk assessments

Identifying human actions that are critical to safety places emphasis on operating activities that, if not performed in accordance with the procedure, could result in a major loss. The nuclear industry, through the Nuclear Regulatory Commission (NRC), handles this category through their human-factors-engineering (HFE) programs. The "major loss" concern includes damage to the nuclear reactor core. The goal of the programs is to minimize the likelihood of human error and ensure that personnel can detect and recover from any errors that may occur.

The NRC conducts a review for every plant-design application to verify that the applicant has

- Identified specific important human actions tied to human factors communication, applied leadership, personal and team development, and risk awareness and mitigation. The impact in safety performance was notable, and from a pilot application, a global implementation was planned (Barakat et al. 2010).

These examples represent the beginning stages of how human-error reducing strategies are effectively being implemented. The obvious next step is to begin the process of integrating these efforts across the industry, not just within individual companies.

4.2 Human/Management System Interaction

The second of the three key relationships is between the worker and the management system (operating or safety). Management systems provide direction and guidance for people, the organization, and contractors on how to operate. When the effectiveness of a management system fails, the results can be catastrophic.

In 1996, 110 people were killed when their DC-9 aircraft, flown by ValuJet, crashed shortly after takeoff from Miami International Airport. The National Transportation Safety Board (NTSB) concluded that a fire had started in the cargo hold of the DC-9, cutting through the flight controls and rendering the aircraft uncontrollable by the flight crew. The fire was linked to hazardous cargo that the contract service company had not properly prepared or presented to ValuJet before it was loaded onto the aircraft. The NTSB determined that contributing to the incident was ValuJet's failure to oversee its contract maintenance program to ensure compliance with maintenance, maintenance training, and hazardous-materials requirements and practices.

In the oil and gas industry, Piper Alpha had a similar effect, setting in motion the evolution of the safety-case and safety-management systems. In 2010, Transocean and the *Deepwater Horizon* management systems were rendered ineffective in preventing or responding to the flow of hydrocarbons in the riser and subsequent explosion and fire. The United States Coast Guard (USCG) accident report stated (USCG 2011):

> *The Safety Management System failed to provide proper risk assessment, adequate maintenance and material condition, and process safety adherence. The Flag State and USCG did not identify these system failures in time to ensure the safety of the vessel.*

These fatal incidents highlight the importance of the relationship between management systems and human factors. Safety-management systems in and of themselves are not the answer, but the discipline of human factors helps to uncover gaps in management systems, helping to ensure their effectiveness.

One of the key elements within a management system is a detailed set of specific and mandated operating procedures. These procedures indicate to the human operator the way the organization intends to have various tasks performed. The intent of operating procedures is to ensure the following and, ultimately, to deliver a predictable outcome:

- Provide guidance to the worker to ensure logical, efficient, safe, and predictable means of carrying out work
- Reduce the variability of work
- Deliver clear expectations
- Define boundaries of acceptability of work

5.0 Achieving a Culture of Perfection Led From the Top

Oil and gas companies will never be "High Reliability Organizations" if they rely on campaigns to change hearts and minds on the operational frontline. Instead, they must identify the obvious precursors to catastrophe and get serious about eliminating them – led firmly from the top.
–Andrew Hopkins, 2014

Zero harm as an expectation will require a cultural adjustment for the industry. Patrick Hudson's Safety Culture Maturity Model (Hudson 2001) is a recognized framework used by organizations to assess and understand their culture. For the 2015–16 SPE Getting to Zero sessions, the Hudson model was adapted to evaluate HSE culture and risk tolerance **(Fig. 4).**

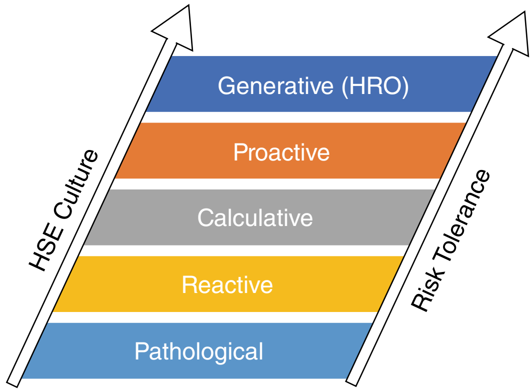

Fig. 4—Culture/risk tolerance adaptation of Hudson's Safety Culture Maturity Model.

Participants at the SPE Getting to Zero sessions identified that an evolution to a generative mindset is emerging within their companies, for both HSE culture and risk tolerance **(Fig 5).** Specifically,

- Two-thirds (69%) identified their organizations as having a proactive/generative HSE cultural mindset: 50% proactive/19% generative.
- Three-quarters (78%) identified their organizations as having a risk-tolerance mindset of being naturally proactive/generative: 72% proactive/6% generative.

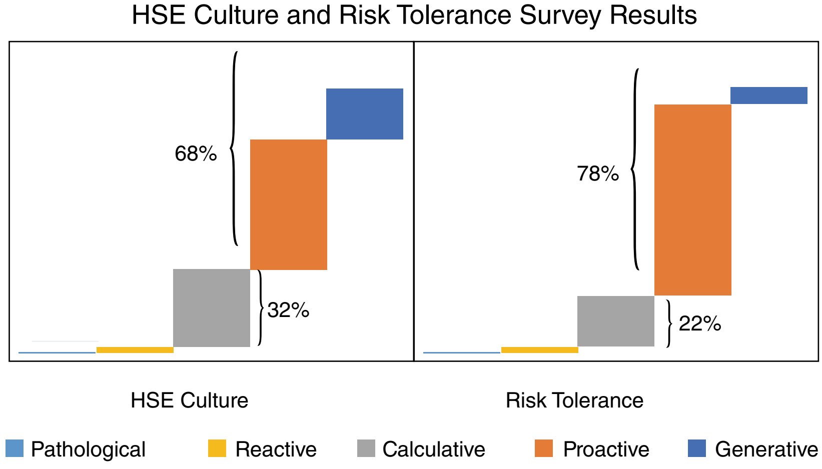

Fig. 5—HSE culture and risk tolerance survey results from SPE Getting to Zero sessions.

Although the more than 750 session participants represented a diverse cross section of the industry, the authors are not concluding that the survey unequivocally defines where the oil and gas industry is on its evolutionary journey to a generative mindset. Rather, the authors attest that the results represent insight into the industry's desire and the direction in which the industry is moving on attributes that are essential for achieving zero as an expectation.

Also indicative of the culture across the industry of the ability to achieve an expectation of zero harm are the present attention and action that are being given to incidents. As a frame of reference, incidents are categorized as fatalities, lost-time injuries, medical cases, first-aid, high-potential, and near-miss. When asked what level of safety incidents trigger their company's incident analysis and attention, the participants reported the following **(Fig. 6):**

- More than one-third of the session participants identified that in their organizations all incidents receive analysis and attention regardless of outcome severity.
- For nearly one-half of the participants, high-potential incidents initiate action in their organizations.
- The small remainder of participants limit their analysis and attention to traditional incident metrics (that is, fatalities, lost time, medical cases, and first aids).

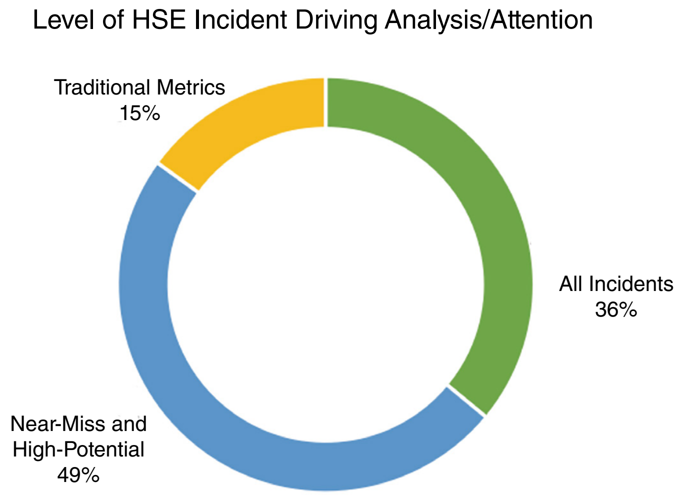

Fig. 6—Management of HSE incidents—survey results from SPE Getting to Zero sessions.

When session participants were queried regarding their company's state of cultural readiness to embrace zero as possible, the results were notable.

- 2% expressed no desire to pursue, and just 16% cited a present lack of leadership commitment or limitations in their present means to manage HSE as what was holding them back.
- 33% indicated it may take some time and clarity as to "how" they would embrace zero as possible.
- Nearly one-half (49%) responded that they are actively working now or are ready to join others in the pursuit.

Most encouraging are the organizational behaviors cited during the SPE Getting to Zero sessions that underpin these positive percentages. More organizations now genuinely believe safety is critical to business success and that their safety systems are essential, using effectiveness methodologies to ensure their efficacy. For such as resource planning and competencies. We will be able to regularly include human-factors assessments in such things as hazard and operability studies/hazard-identification studies and equivalent tools/processes so we can identify shortcomings in the understanding of the engineered process. Incident investigations and root-cause analysis can include a common set of human-factors attributes that would allow companies to better understand contributing factors in events that go well beyond the often-cited and yet narrow assessment that someone did not follow process or that the event was caused by operator error.

Industry conferences and technical forums provide the opportunity to further the discussion and share broader ideas and actions started by the industry. Examples of these opportunities include

- 2014 SPE Workshop Getting to Grips With Human Factors in Drilling Operations: The main points from the workshop presentations and discussions were summarized in a paper that included practical tools and insights that could be applied to day-to-day operations and additional resources for further reading (Thorogood et al. 2015).
- A panel discussion at the 2017 Offshore Technology Conference included a presentation by a service company about their work in human factors. Their efforts are focused on the underlying organizational and system aspects that support the individual. Using their "What Lies Beneath" approach **(Fig. 7)**, which incorporates human factors, they are embracing a holistic view that is enabling their organization to achieve a step change, not just incremental change, in HSE performance. This will further their cultural advancement toward an HRO (Hinton 2017).

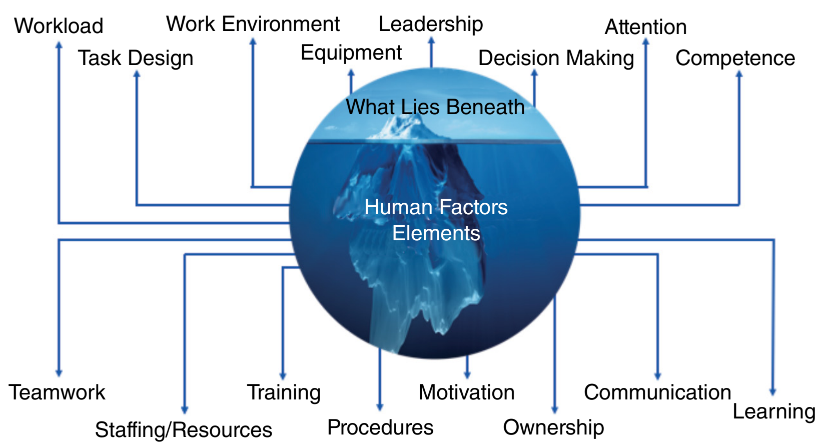

Fig. 7—What-lies-beneath analysis of human-factors elements (Harris and Hinton 2017).

3. **De-emphasize lagging performance indicators and use leading indicators.** As an industry, we have become adept at gathering performance on lagging indicators (events that have occurred) vs. leading indicators (risk-control measures). It is easy to measure safety by counting injuries, but much more complicated to measure (and set targets for) effective controls to reduce or eliminate risk. While the emphasis on lagging indicators has brought significant improvement in recordable injury rates, the pressure exerted by focusing on incidents as a performance measure is yielding diminishing returns, to the point that incident rates are no longer a reliable indicator of a company's safety program (OSHA 2012).

   Goodhart's Law applies here: "Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes" (Goodhart 1975). Or in other words, "when a measure becomes a target, it can no longer be used as a measure" (Strathern 1997).

of company, and be assured of support from the facility owner (operating company representative). Of course, the ideal state is not the real state at many oil and gas worksites.

a. Interdependent HSE culture. If a company has a mature interdependent HSE culture, its employees are more likely to recognize and act on hazardous conditions. The DuPont Bradley HSE culture model **(Fig. 8)** describes the journey to an HSE culture in three phases, moving from a dependent phase, through independence, to an interdependent culture. In the dependent phase, HSE is mainly driven through use of control, discipline, rules, and regulations. As the culture matures, it moves through the independent phase, in which employees begin to take personal responsibility for HSE rather than simply relying on rules and regulations to create a safe work environment. The final—interdependent—phase is characterized by a "peer's keeper" approach that is adopted by all. In this phase, employees do not just look out for their own safety; everyone looks out for each other's safety, and management works collaboratively with employees on HSE matters, feeling comfortable leading or allowing others to lead. People do this because they genuinely care about the safety and well-being of their colleagues. They do it because they want to, not because they have to.

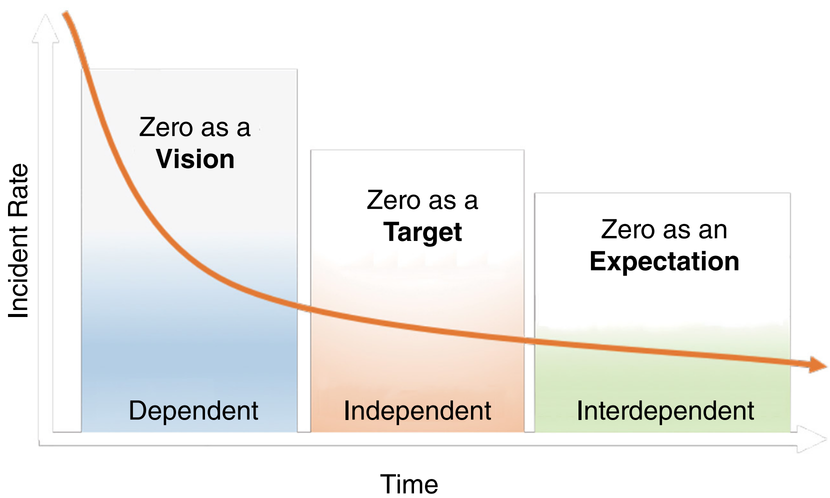

Fig. 8—DuPont Bradley HSE culture modified to include zero.

b. Operational ownership of HSE. Interdependence is established by achieving operational ownership of HSE—that is, by employees assuming accountability for HSE performance and maintaining a safe work environment. Operational HSE ownership is characterized by employees looking out for each other, intervening in unsafe acts and conditions, and engaging with HSE to identify and implement risk-mitigating controls and processes. This leads to a collaborative engagement rather than HSE directing and acting as a police force in the organization. Operational ownership, however, must exist at every level of the organization to be successful.

c. Sustainable HSE leadership. While operational ownership of HSE leads to all employees assuming accountability for HSE performance, it is essential that senior leaders and managers be a continual driving force in embedding this concept throughout the organization. Senior leadership and managers have a direct effect on establishing the "culture" of the organization and on employee HSE behaviors in the workplace. The behavior of managers, through their influence on employees, can strongly

7.0 Appendices

7.1 Acronyms

API	American Petroleum Institute
BSEE	Bureau of Safety and Environmental Enforcement
BTS	Bureau of Transportation Statistics
COS	Center for Offshore Safety
CRM	Crew Resource Management
DOD	Department of Defense
DOE	Department of Energy
FAA	Federal Aviation Administration
HFE	Human-Factors Engineering
HRA	Human-Reliability Analysis
HRO	High-Reliability Organization
HSE	Health, Safety, and Environment
HSSE-SR	Health, Safety, Security, Environment, and Social Responsibility
IADC	International Association of Drilling Contractors
IAGC	International Association of Geophysical Contractors
IOGP	International Association of Oil and Gas Producers
NASA	National Aeronautics and Space Administration
NRC	Nuclear Regulatory Commission
NTSB	National Transportation Safety Board
SPE	Society of Petroleum Engineers
USCG	United States Coast Guard

7.2 Participant Demographics for SPE Getting to Zero: The Road to Stavanger Sessions

SPE Getting to Zero: The Road to Stavanger sessions conducted across the globe were attended by more than 750 participants. During the interactive sessions, participants were asked to report their organizational affiliation and their employment position. The results are illustrated by the graphs below.

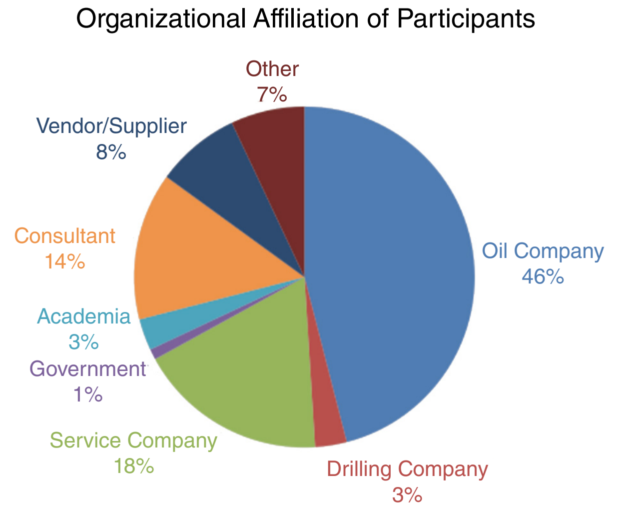

Fig. 9—Organizational affiliation of participants of the SPE Getting to Zero sessions.

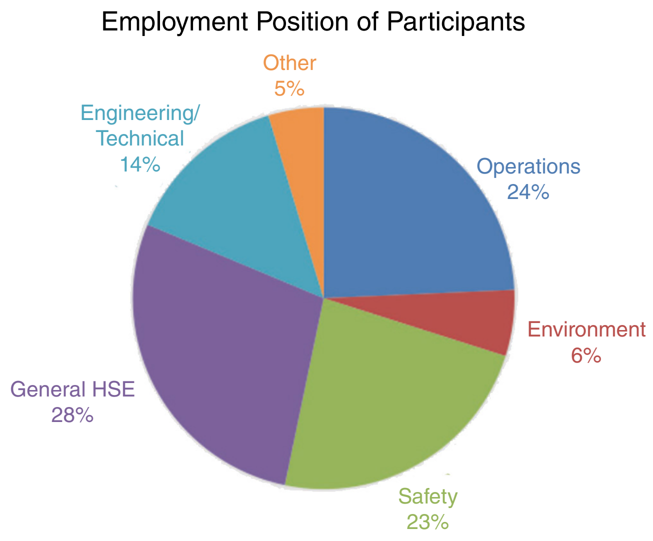

Fig. 10—Employment positions of participants of the SPE Getting to Zero sessions.

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SPE Technical Report

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Richardson, TX 75080-2040
972-952-9393
www.SPE.org
2018

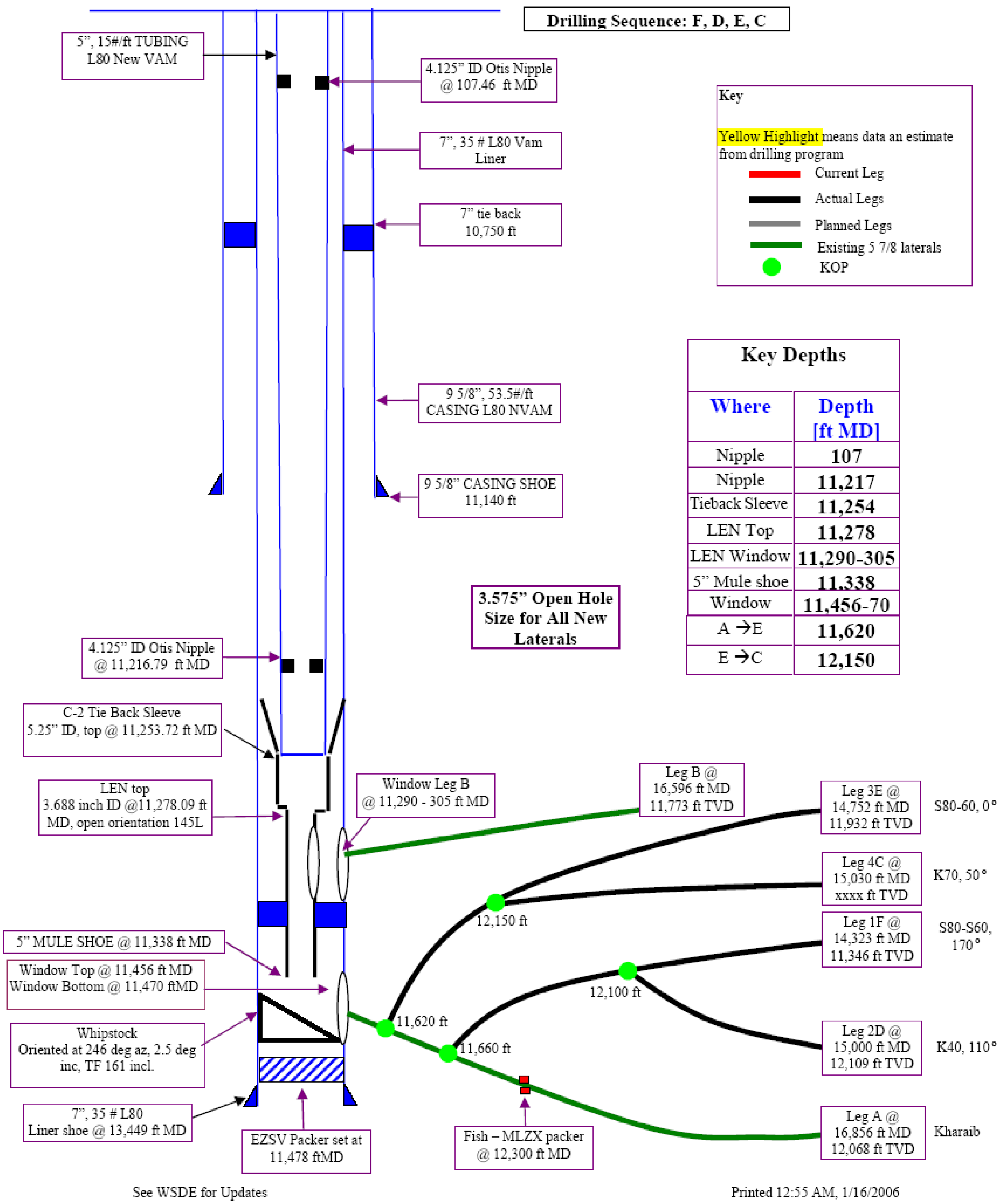
SAJAA 9 Coil Tubing Underbalanced Drilling

January 2006

End of Well Report

Revision:	2
Date:	23rd February 2006
Prepared by:	R. Pickles
	J. Rennox
	T. Kavanagh
	P. Savage
Prepared for:	bp Sharjah Oil Company

Wellbore Diagram Sajaa 9



Key Depths

Where	Depth [ft MD]
Nipple	107
Nipple	11,217
Tieback Sleeve	11,254
LEN Top	11,278
LEN Window	11,290-305
5" Mule shoe	11,338
Window	11,456-70
A →E	11,620
E →C	12,150

Sajaa 9 - Coiled Tubing Acid Job (Summary) - 1998

The coiled tubing acid job work was done in February 1998. The purpose of the work was to remove OBM and stimulate both laterals as well as testing the re-entry capability of the Baker Lateral Entry Nipple (LEN) system.

From the IP standpoint this has been ASOC's most successful coil tubing stimulation to date. Possible reasons include concentrating the acid in the porous and naturally fractured intervals, acidizing the well while SI which allows the spent acid head to help "squeeze" the acid into the target zones. A 4 to 5 MMCFD initial production increase was realized.

Summary of runs:
9a

- 2-3/8" dummy run to 14,813' flooded with diesel and ran to 15,971' with stick/slip last 1100'
- After flowing to recover diesel and spotting xylene CT became stuck
- Pulling and working pipe unsuccessful, pumped 100 BBL acid and all pulled free
- Acidized 15,600' - 14,800' , 13,700' - 11,850'
- Cleanup recovered mud/acid insolubles

9b

- Len ID too small for 2-3/8" CT - with diverter installed
- Installed CT diverter on slick line
- RIH and stacked out on something
- Due to unresolved technical challenges, aborted job
- "A' acid job increased productivity by + 5 MMCFD

Sharjah Historical Intervention Tracking sheet

<u>Well: Sajaa 09</u>

<u>11.Feb.98</u>	CT RIH with 2.3/8" coil to 15,890ft. (CT) in leg A and pump xylene and foamed acid.
<u>15.Feb.98</u>	W/L ran whipstock and set in LEN.
<u>16.Feb.98</u>	CT make dummy run with 2" CT into upper leg B. Hung up at 11,307ft (CT). Abandoned run. Wireline pulled whipstock.
<u>06.Nov.01</u>	As a precaution, cellar dug out, shims installed to support 20" csg. and cellar re-concreted.
<u>17.Nov.02</u>	Check valve and choke replaced with manual gate valve and flow tee.
<u>18.Nov.02</u>	Ran 3.42" GC to 11,512'rkb. No obstructions or anomalies found.
<u>25.Nov.02</u>	Ran PSP to 11,450ftrkb HUD. Flow was 72% from Shuaiba and 28% from Kharaib. During SI pass cross-flow was shown from the Shuaiba to the Kharaib. Max. BHT was 276 degrees F.
<u>29.Nov.05</u>	Ran 3.6" GC to 11,287ft rkb. Ran 2.5" GC to 11,511ft rkb. Couldn't get 2.75" GC passed 11,294ft rkb.

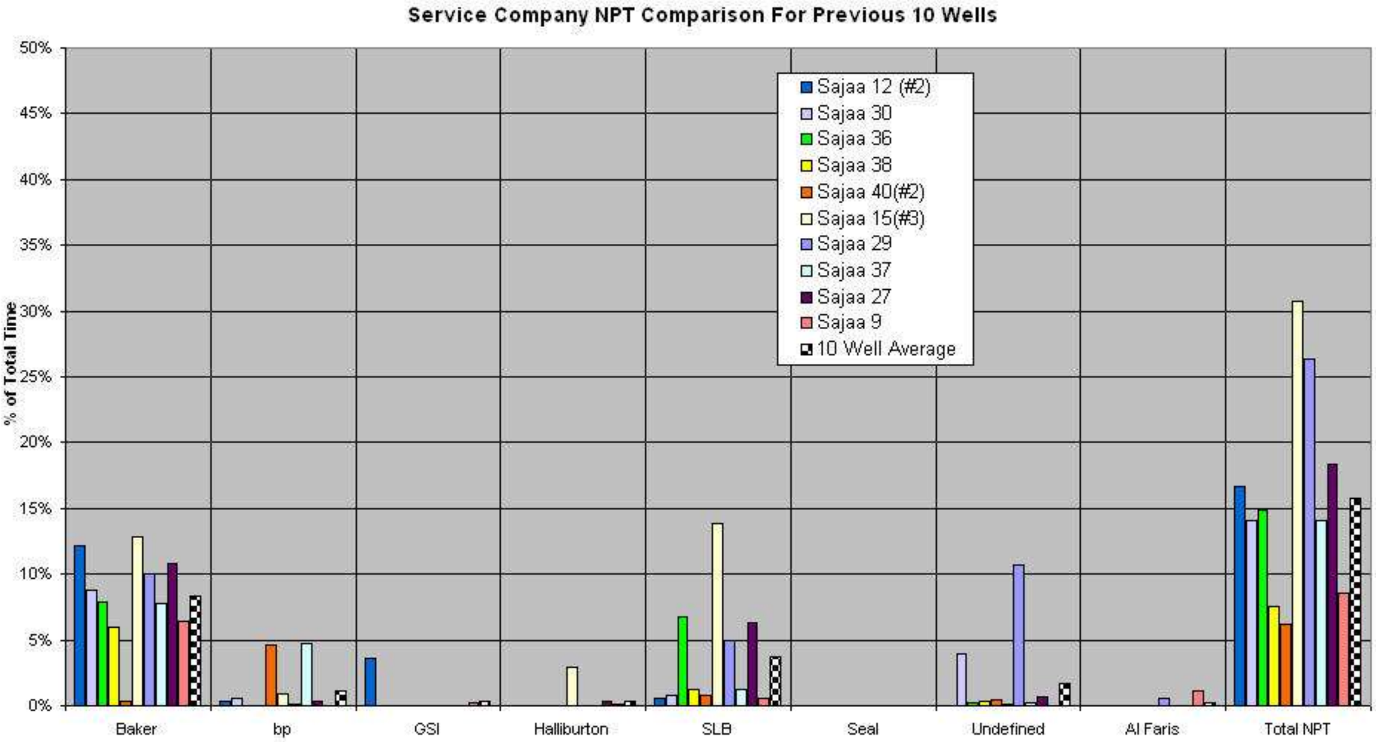
NPT Analysis

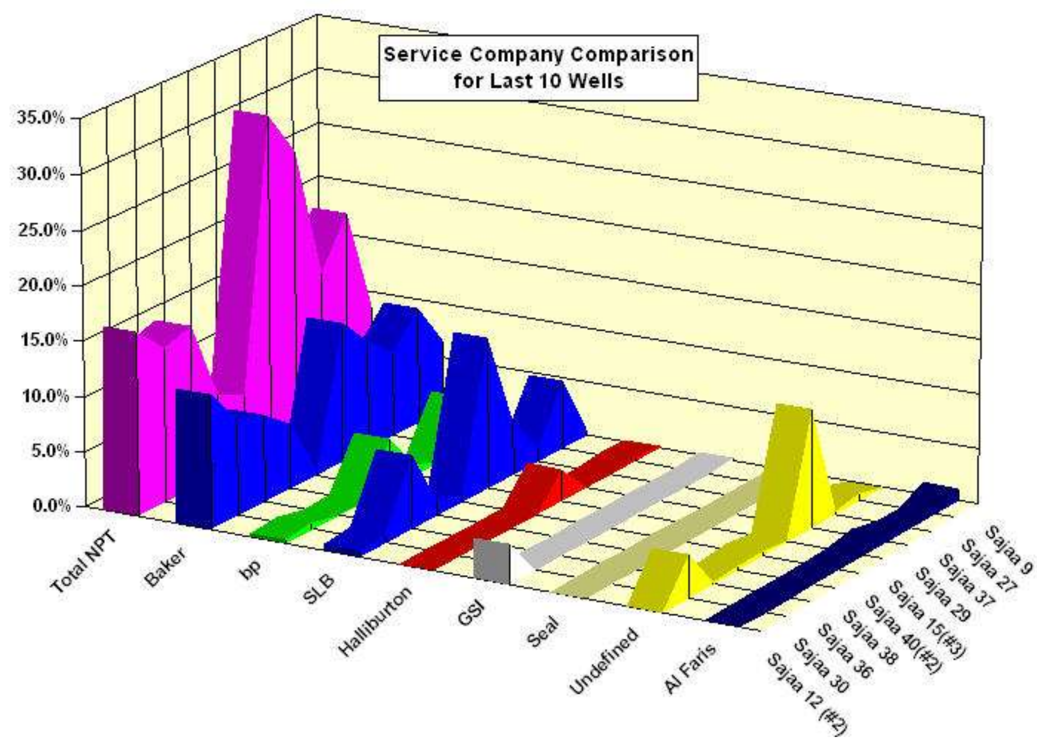
NPT for Sajaa 9 was 8.2% of total time and 5.1% of total cost. This is the 3rd lowest NPT time and 4th lowest NPT cost since the start of the project.

Totals	Time [Hours]	Cost	% of Total Time	% of Total Cost
Baker	27.40	$131,424	6.3%	4.4%
bp	0.00	$0	0.0%	0.0%
GSI	1.00	$4,136	0.2%	0.1%
Halliburton	0.58	$2,399	0.1%	0.1%
Schlumberger	2.16	$12,909	0.5%	0.4%
SEAL	0.00	$0	0.0%	0.0%
Misc/Weather	0.00	$0	0.0%	0.0%
Al Faris	4.17	$2,710	1.0%	0.1%
Total NPT	35.31	$153,578	8.2%	5.2%
Total Well	432.00	$2,971,485		
Productive	396.69	$2,820,617	91.8%	94.9%
Rig Days	18.0	Days		

Every failure of a BHA was recorded as NPT against Baker (NB: this includes BHI & BOT) as per bp standard procedure, to remove any subjectivity from the appliance of NPT criteria. However, bp considers the use of 2 milling BHA's per window (1 to run Whipstock and one to mill the window) and 1 drilling BHA per leg as a reasonable technical limit for this project, given the down hole temperatures and drilling fluid pumped. Accordingly, the NPT for Baker using these criteria would have been reduced to:

Totals	Time [Hours]	Cost	% of Total Time	% of Total Cost
Baker	**Totals**	**Time [Hours]**	**Cost**	**% of Total Time**
bp	Baker	**3.9**	**$18,775**	0.9%
GSI	bp	0.0	$0	0.0%
Halliburton	GSI	1.0	$4,136	0.2%
Schlumberger	Halliburton	0.6	$2,399	0.1%
SEAL	Schlumberger	2.2	$12,909	0.5%
Undefined	SEAL	0.0	$0	0.0%
Al Faris	Undefined	0.0	$0	0.0%
Total NPT	Al Faris	4.2	$0	1.0%
Total Well	Total NPT	7.7	$38,219	1.8%
Productive	Total Well	432.0	$2,971,485	


Service Company NPT Comparison For Previous 10 Wells
% of Total Time
50%
45%
40%
35%
30%
25%
20%
15%
10%
5%
0%
Sajaa 12 (#2)
Sajaa 30
Sajaa 36
Sajaa 38
Sajaa 40(#2)
Sajaa 15(#3)
Sajaa 29
Sajaa 37
Sajaa 27
Sajaa 9
10 Well Average
Baker
bp
GSI
Halliburton
SLB
Seal
Undefined
Al Faris
Total NPT


Service Company Comparison for Last 10 Wells
35.0%
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
Total NPT
Baker
bp
SLB
Halliburton
GSI
Seal
Undefined
Al Faris
Sajaa 12 (#2)
Sajaa 30
Sajaa 36
Sajaa 38
Sajaa 40(#2)
Sajaa 15(#3)
Sajaa 29
Sajaa 37
Sajaa 27
Sajaa 9

NPT Time Breakdown:

BAKER				
Date	Time	Problem	Daily Cost	NPT cost
11-Jan	14.07	Orienter failure	$106,441	$62,401
11-Jan	6.91	GR tool failure	$106,441	$30,646
12-Jan	6.00	Orienter failure	$143,465	$35,866
12-Jan	0.42	Re-boot BHI computer	$143,465	$2,511
Totals	**27.40**		**Dollars**	**$131,424**

BP				
Date	Time	Problem	Daily Cost	NPT cost
Totals	**0.00**		**Dollars**	**$0**

SLB				
Date	Time	Problem	Daily Cost	NPT cost
01-Jan	1.24	Volvo Powerpack air compressor head gasket blown - replace.	$134,955	$6,973
09-Jan	0.50	CCAT crash.	$154,861	$3,226
12-Jan	0.42	Friction wheel slipping, switch to UTLM and re-log section.	$154,861	$2,710
Totals	**2.16**		**Dollars**	**$12,909**

GSI				
Date	Time	Problem	Daily Cost	NPT cost
14-Jan	1.00	Manual dump valve from scrubber opened in error.	$99,264	$4,136
Totals	**1.00**		**Dollars**	**$4,136**

HES				
Date	**Time**	**Problem**	**Daily Cost**	**NPT cost**
14-Jan	0.58	Condensate pump problems	$99,264	$2,399
Totals	**0.58**		**Dollars**	**$2,399**

Seal				
Date	**Time**	**Problem**	**Daily Cost**	**NPT cost**
				$0
Totals	**0.00**		**Dollars**	**$0**

Misc/Weather				
Date	**Time**	**Problem**	**Daily Cost**	**NPT cost**
				$0
Totals	**0.00**		**Dollars**	**$0**

Al Faris				
Date	**Time**	**Problem**	**Daily Cost**	**NPT cost**

Appendices

Appendix 1: HSE Statistics

Environmental Totals

Gas Compressed (mmscf)	Hydrocarbon Gas Flared	Cold Vent (mscf)	Condensate To Plant [bbl]	Water To Land Farm (bbl)	Water Pumped to Formation (bbl)
58.489	9.005	51.60	531.7	9,475.6	-219.1

12.60% Hydrocarbon gas flared on this well.

23,994 man-hours worked on Sajaa #9 without a DAFWC.

1,193,143 man-hours without a DAFWC for the entire project thus far.

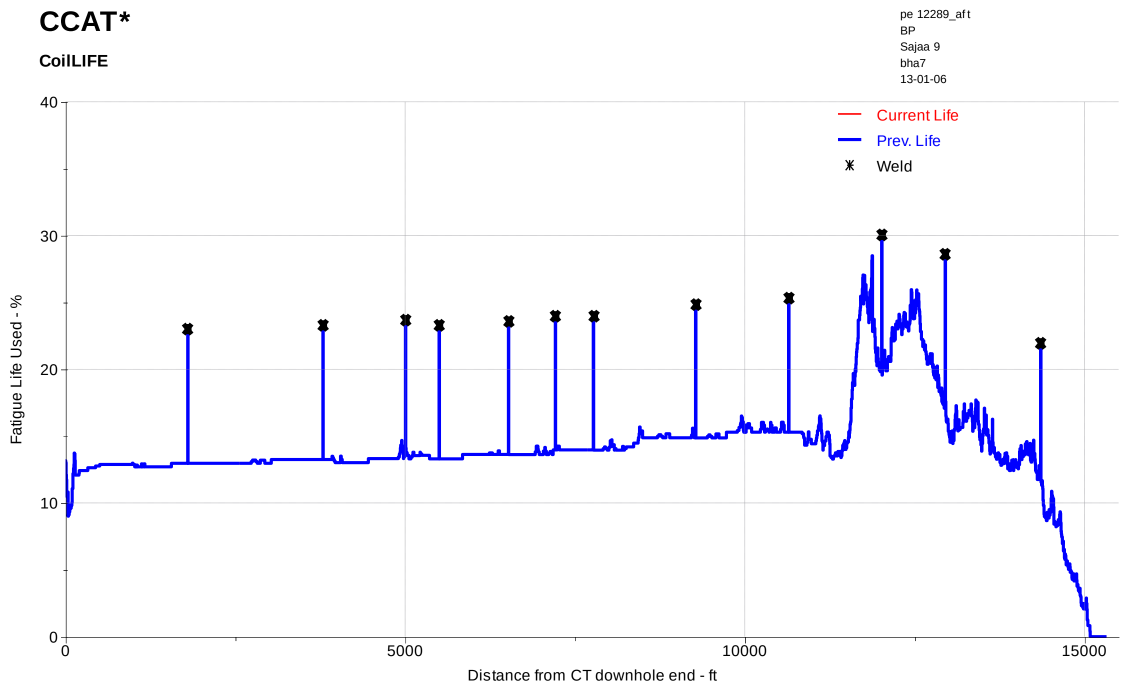
Appendix 2: Services Provided by Contractors

Wellsite Engineering	Blade Energy, Schlumberger, Halliburton
Wellsite Supervision:	BP, Blade Energy
BHA's, ADMs, and Directional:	Baker Hughes Inteq
Whipstock, Fishing Tools, PDMs:	Baker Oil Tools
Generator, Light Plants:	Caterpillar
Cranes, Construction Equipment:	Al Faris
Gas Compression:	GSI (Weatherford)
UBD Returns Package:	Halliburton
Materials Fabrication, Labour:	MIS, Lamprell
Geosteering Services:	Robertson Research
Coiled Tubing:	Schlumberger
Nitrogen:	Schlumberger
Paramedic, Safety, Gas Detection:	Seal International

Window Top @ 11,456 ft MD Window Bottom @ 11,470 ftMD C-2 Tie Back Sleeve 5.25" ID, top @

Job #	Job Date	Job ID	Client Name	Well Name	Job Description	GN ϕ (in)	H_2S Level (ppm)	CO_2 (%)
18	11-Jan-06	BHA 6	BP	Sajaa 09	Drill Leg 4C - Fail	100	40	4
19	11-Jan-06	BHA 7	BP	Sajaa 09	Drill Leg 4C	100	40	4
20	14-Jan-06		BP	Sajaa 09	cut 15ft CT & 15 cable			

Coiled Tubing Fatigue Life Plot for PE 12289



HROs have procedures that are designed to prevent errors, are standardized in format, and function to ensure that they are followed by workers throughout their industries, regardless of company affiliation. Performance mode thinking, as an aspect of human factors, helps to create procedures that consider the factors that produce errors and attempts to eliminate them when procedures are created.

In the oil and gas industry, differences exist in contractors' operating procedures that sometimes exhibit little consistency. Such nonstandardized work processes lead to deviations by individuals and crews and difficulty in training people as they move from one location or company to another. These deviations continue to occur in the oil and gas industry and are typically masked on a contractor's mishap report under the label "not following procedures," which does not cite proper corrective actions other than "follow the procedures" to prevent similar incidents. Furthermore, the more routine the work becomes for a worker, the more apt the worker is to drift from the original/mentored way of conducting work. This effect is known as procedural drift, an adaptation of practical drift (Snook 2000), the search to establish why formal safety procedures may not ensure system safety.

There is a concern that the industry is exposed to workers who learn their craft and reapply knowledge from each other more than from formal/procedural training (Caldwell and Hinton 2015). This approach to conducting work has left a large variable in the output of the work and highlights the development of workers as the single point of failure in the management system.

The nuclear and commercial aviation industries provide an interesting counterpoint to this approach. These industries developed detailed, standardized, step-by-step operating procedures to help prevent human error during routine work, where most incidents occur in the oil and gas industry. These highly detailed operating procedures became the foundation of the safety or operational management system for each organization in their industry and created a step change in safety. Today, operating procedures are the cornerstone of any corporate management system throughout these industries because they are used for training, investigation, quality control, auditing, continuous improvement, and organizational learning.

There are efforts underway within the oil and gas industry to change the course. An oil and gas service provider designed a system of checklists for operating perforating guns with a goal of lowering their misfire rate. Although the project was initially identified as an operational enhancement and not specifically a safety improvement, the focus on risk (misfire) reduction provided improvements in safety, reliability, and operations. Perforating-gun checklists were designed to catch human errors during the process of rebuilding the guns. Three months after crews were trained on how to use the checklists and with more than 1,600 gun runs, the results were staggering. With a measured operational improvement of 74% within a few months and a quarterly savings of more than USD 18 million, the service provider declared the project a tremendous success. Three years later, the teams that received the checklists and training realized reliability improvements of more than 300% (Dingee 2015).

The importance of the worker to their management system cannot be overstated. Company management systems in the oil and gas industry have been in a constant state of overhaul and revision for the last 20 years to improve safety culture and organizational performance. This is because humans govern and accomplish all the activities necessary to control the risk of incidents. Not only do humans unintentionally make errors in executing a process that can result in an incident, they also make errors by creating deficiencies in the design and implementation of management systems. We are prone to making errors in all areas of management systems such as authorities, accountabilities, procedures, feedback, proof documents, and continual